

claude-windows / 6e9d0958-a2ec-4a8e-bb2d-6cce9833c527

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\subagents\agent-ae9ca45a9b9abd0b4.jsonl`
- SHA-256: `25a55a6f9c95c901dc69bceb540c8cf45475c20d7acad4595616a845406c6b24`
- Source modified: `2026-07-04T00:46:08+00:00`
- Imported at: `2026-07-05T16:48:26+00:00`
- project: `subagents`
- session_id: `6e9d0958-a2ec-4a8e-bb2d-6cce9833c527`

Transcript

1. user (2026-07-04T00:45:16.564Z)

Code-review finder. Target: PR #458 of badminton-highlight-indexer. Diff: `cd C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer && gh pr diff 458`. PR branch = `origin/codex/a7-a9-quality-chain`. Changed files: backend/eval/fusion_golden.py, backend/eval/fusion_audio.py, tests/test_ablation.py, tests/test_audio_features.py, tests/test_fusion_compare.py, docs/ALPHA_LAUNCH_READINESS.md, SESSION_HANDOFF.md.

ANGLE ? conventions (CLAUDE.md). Read the governing CLAUDE.md files:

C:/Users/avidu/.claude/CLAUDE.md (user global), the repo-root CLAUDE.md and any CLAUDE.md in backend/, backend/eval/, docs/, tests/ that are ancestors of changed files. Also read docs/PROJECT_DOC_TEMPLATE.md §7 legend + the ALPHA_LAUNCH_READINESS.md honesty-tag conventions ([verified]/[analysis]). Check the diff for CLEAR violations ? quote the exact rule and the exact violating line:

1. Honesty tags: the doc adds many `[analysis]` claims (A7/A8/A9 results, F1 numbers). Are load-bearing quantitative claims properly tagged per the doc's own honesty-note convention? Any claim stated as fact that should be [analysis]?
2. The tracker marks A6 DONE (#455), A3 IN PROGRESS (collector #62), A7/A8/A9 IN PROGRESS (this PR). The PROJECT_DOC_TEMPLATE §7 legend defines these states ? is any status inconsistent (e.g. a row marked DONE whose PR isn't merged, or a "this PR" reference on a DRAFT PR)?
3. Attribution rule (working-agreement-attribution): does the doc assert any provenance/result that didn't happen? The results reference specific artifact paths (artifacts/rally_tal_gen0/0.1.1-n15-shadow.20260704) ? the PR body says generated artifacts are NOT committed, so are these referenced-but-absent artifacts a broken reference per doc conventions?
4. SESSION_HANDOFF.md is edited ? the global CLAUDE.md session-handoff convention says it reflects real shipped state; does the handoff claim anything aspirational (e.g. "A6 PR #455 is merged" ? verify #455 is actually merged)?

Only flag with an exact quoted rule + line. If no CLAUDE.md rule is broken, return []. Return up to 6 candidates as JSON [{file, line, summary, failure_scenario}]. Final message = only the JSON array.

2. assistant (2026-07-04T00:45:18.450Z)

I'll help you review PR #458 for convention violations. Let me start by gathering the necessary context files and the diff.

3. user (2026-07-04T00:45:20.893Z)

<persisted-output>

Output too large (51.9KB). Full output saved to: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\tool-results\bxvgydfjh.txt

Preview (first 2KB):

```
diff --git a/SESSION_HANDOFF.md b/SESSION_HANDOFF.md
index 31ea4c14..a50cc8a1 100644
```

```

--- a/SESSION_HANDOFF.md
+++ b/SESSION_HANDOFF.md
@@ -2,49 +2,51 @@

## You are here

-A6 is implemented and open for review as PR #455:
-`https://github.com/Khelsutra/badminton-highlight-indexer/pull/455`.
+A6 PR #455 is merged. This checkout is on `codex/a7-a9-quality-chain`, fast-forwarded to
+`origin/master` at `c5fc27c`, with tracked edits in:

-The branch is `codex/a6-ablation-cli`, rebased onto `origin/master` after A5
-merged as `c8eec7c` / PR #454. It fixes the `python -m backend.eval.ablation`
-circular import by making `ablation.export_pdf` a lazy wrapper, adds a
-module-entrypoint regression test, and records the n=15 R2/tolF1 contrast in
-`docs/ALPHA_LAUNCH_READINESS.md`.
+- `backend/eval/fusion_golden.py`
+- `backend/eval/fusion_audio.py`
+- `tests/test_fusion_compare.py`
+- `tests/test_audio_features.py`
+- `docs/ALPHA_LAUNCH_READINESS.md`
+- `SESSION_HANDOFF.md`

-Validation completed locally on 2026-07-03:
-`ruff check .`, `python -m mypy backend training`, and
-`python run_all_tests.py --cov=backend --cov=training --cov-report=term --cov-fail-under=70`
-all passed. Full suite result: 1609 passed, 6 skipped, 85.65% coverage.
+A7/A8/A9 have been run locally against the 15-video corpus, and draft PR #458 is open:
+`https://github.com/Khelsutra/badminton-highlight-indexer/pull/458`. The generated
+feature/model artifacts live under ignored `output/` and `artifacts/` paths and should not
+be committed.

## Next steps / open threads

-1. Review/merge PR #455 for A6.
-2. A7 is the next indexer-owned quality item: regenerate or upconvert
-   full-fusion feature JSONs for the newer nine golden videos so fusion/person
-   /audio analyses can run on all 15 videos. GPU is approved on this machine.
-3. A8 follows A7: rerun `fusion_compare`, group ablation, and served contrast;
-   explicitly promote or reject fusion/person/Gate-A served-default decisions.
...
</persisted-output>

```

4. user (2026-07-04T00:45:20.929Z)

```

1      # Global instructions (all projects)
2
3      ## Repo-freshness convention: git pull BEFORE reading
4      **Always `git pull` a repo before starting to read it** ? at session start for the
primary
5      working directory, and again for any sibling/local repo the moment work touches it.
6
7      - **Why:** the owner closes every session on a "clean slate", but multiple parallel
slates
8      (sessions/machines) exist ? PRs get merged and master moves between sessions, so the
local
9      checkout can silently lag the remote truth. Pull first so ramp-up reads (handoff,
docs, code)
10     reflect where the remote actually is.
11     - **How (safe form):** `git pull --ff-only` on the current branch (a plain fetch+ff).
Never
12     auto-merge, rebase, or discard local state to make a pull succeed.
13     - **If it can't fast-forward** (diverged branch, dirty tree in the way): do NOT force
anything ?
14     report the state to the owner and proceed read-only on what's local, flagging that it

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may be
15     stale. Uncommitted changes are often a sibling session's or the owner's WIP; never
clobber them.
16
17     ## Git branching safety: concurrent sessions / shared checkouts
18     Multiple sessions ? and spawned/background tasks ? may share ONE working checkout and
create
19     branches + commit **concurrently**, so `git branch` / `git log` can change UNDER you
between two
20     commands. (Spawned "fresh worktree" tasks are NOT always isolated.) This has caused a
real
21     incident: a sibling commit landed in the gap between a branch-point check and `git
checkout -b`,
22     so the new branch rode on top of it and a **squash-merge silently absorbed the sibling's
work**.
23     Protocol ? do this EVERY time, not only when you suspect a collision:
24     - **Branch from the REMOTE, explicitly:** `git fetch origin` then `git checkout -b
<name> origin/<base>`
25     (e.g. `origin/master`). Never assume the current branch is the intended base.
26     - **Re-verify right before `git add`/commit:** `git branch --show-current` + `git log
--oneline -1`
27     (HEAD is yours). Before opening a PR, `git log --oneline origin/<base>..HEAD` must
list ONLY your
28     commits ? if a sibling commit appears, re-cut the branch from `origin/<base>`.
29     - **Stage explicit paths** (`git add <files>`) ? never `git add -A`/`.`/`-u`, so sibling
or untracked
30     files can't ride into your commit.
31     - **Serialize repo writes:** don't start a repo-writing spawned/background task while
you hold
32     uncommitted work ? commit or push first. If one may be running, treat the tree +
current branch
33     as able to shift, and put this branching-safety reminder into the spawned task's own
prompt.
34
35     ## Session-handoff convention
36     Maintain a living **session handoff** for each project and use it to resume context
quickly
37     across windows.
38
39     - **Where it lives:**
40     - If the project has an auto-memory dir (`~/.claude/projects/<project>/memory/`), keep
the
41     handoff as `session-handoff.md` there, and list it under a **"? Start here"*** entry
at the
42     top of that project's `MEMORY.md`.
43     - Otherwise, keep a `SESSION_HANDOFF.md` at the repo root.
44     - **What it contains** (keep it to ~1 page ? it POINTS to detailed artifacts, never
duplicates them):
45     1. **You are here** ? current state.
46     2. **Next steps / open threads.**
47     3. **Ramp-up kit** ? the exact files/docs to read to get current.
48     4. **Key decisions** ? so they aren't relitigated.
49     - **At session start:** if a handoff exists, read it before starting substantive work,
and use it
50     (plus its ramp-up kit) to get up to speed instead of replaying past conversation.
51     - **Updating it:** keep it current **automatically** ? refresh at meaningful checkpoints
(a PR is
52     pushed/merged, a key decision is made, work shifts phase, or a session is wrapping
up), so a
53     restart can ramp up without losing context. Do **not** rewrite it every turn ? only
when
54     something worth resuming from changed. The phrases **"update the handoff"*** /
**"refresh the
55     handoff"*** force an immediate full rewrite on demand.
56     - **Keep it honest:** it reflects real, shipped state ? never aspirational. (See each

```

```

project's
57     attribution/working-agreement note where present.)
58
59     ## Code review ? pre-LGTM gate (applies to every PR/code review, every project)
60
61     Do **not** post **LGTM** or otherwise sign off on a pull request until I have personally
**verified** the following ? never trust the PR description's claims, reproduce them locally:
62
63     1. **Full test suite is green.** Run the whole suite (e.g. `run_all_tests.py` /
`pytest`), not just the changed files.
64     2. **Install any dependency needed** so the suite actually collects and runs ? do not
let modules skip or error for missing deps. (If a check genuinely can't run locally ?
GPU/WSL/real-video/etc. ? say so explicitly and note what was and wasn't verified.)
65     3. **Lint + type checks are clean** ? `ruff check .` and `mypy ...`, matching the repo's
CI. Install `ruff`/`mypy` if absent.
66     4. **Coverage stays up** ? run with coverage and confirm it stays **at or above the
repo's floor** (e.g. the CI `--cov-fail-under` value) and does **not regress** versus the base
branch.
67     5. **Review findings are resolved** ? bugs and repo-convention/discipline issues
addressed.
68
69     LGTM is a *verified* stamp, never a rubber stamp. State plainly what was and wasn't
verifiable locally.
70

```

5. user (2026-07-04T00:45:20.997Z)

CLAUDE.md

6. assistant (2026-07-04T00:45:22.917Z)

Let me read the full diff and the CLAUDE.md file in the repo root:

7. user (2026-07-04T00:45:24.752Z)

```

1     diff --git a/SESSION_HANDOFF.md b/SESSION_HANDOFF.md
2     index 31ea4c14..a50cc8a1 100644
3     --- a/SESSION_HANDOFF.md
4     +++ b/SESSION_HANDOFF.md
5     @@ -2,49 +2,51 @@
6
7     ## You are here
8
9     -A6 is implemented and open for review as PR #455:
10    -`https://github.com/Khelsutra/badminton-highlight-indexer/pull/455`.
11    +A6 PR #455 is merged. This checkout is on `codex/a7-a9-quality-chain`, fast-forwarded
to
12    +`origin/master` at `c5fc27c`, with tracked edits in:
13
14    -The branch is `codex/a6-ablation-cli`, rebased onto `origin/master` after A5
15    -merged as `c8eec7c` / PR #454. It fixes the `python -m backend.eval.ablation`
16    -circular import by making `ablation.export_pdf` a lazy wrapper, adds a
17    -module-entrypoint regression test, and records the n=15 R2/tolF1 contrast in
18    -`docs/ALPHA_LAUNCH_READINESS.md`.
19    +- `backend/eval/fusion_golden.py`
20    +- `backend/eval/fusion_audio.py`
21    +- `tests/test_fusion_compare.py`
22    +- `tests/test_audio_features.py`
23    +- `docs/ALPHA_LAUNCH_READINESS.md`
24    +- `SESSION_HANDOFF.md`
25
26    -Validation completed locally on 2026-07-03:
27    -`ruff check .`, `python -m mypy backend training`, and
28    -`python run_all_tests.py --cov=backend --cov=training --cov-report=term --cov-fail-

```

```

under=70`
29     -all passed. Full suite result: 1609 passed, 6 skipped, 85.65% coverage.
30     +A7/A8/A9 have been run locally against the 15-video corpus, and draft PR #458 is open:
31     +`https://github.com/Khelsutra/badminton-highlight-indexer/pull/458`. The generated
32     +feature/model artifacts live under ignored `output/` and `artifacts/` paths and should
not
33     +be committed.
34
35     ## Next steps / open threads
36
37     -1. Review/merge PR #455 for A6.
38     -2. A7 is the next indexer-owned quality item: regenerate or upconvert
39     -   full-fusion feature JSONs for the newer nine golden videos so fusion/person
40     -   /audio analyses can run on all 15 videos. GPU is approved on this machine.
41     -3. A8 follows A7: rerun `fusion_compare`, group ablation, and served contrast;
42     -   explicitly promote or reject fusion/person/Gate-A served-default decisions.
43     -4. A3 is owned by another agent; do not duplicate that collector/vault work.
44     +1. Address review comments on draft PR #458; convert to ready only after owner/reviewer
agrees
45     +   the A9 TemporalMaxer scope question is settled.
46     +2. Decide whether A9 is considered complete with the current Gen-0 non-serving shadow
47     +   bundle or whether TemporalMaxer needs a follow-up scorer-swap PR.
48     +3. Do not touch A1; another agent owns it.
49     +4. A3 is in collector PR #62 and records all 15 source videos; the two Boxhill
originals
50     +   were verified on `F:\`.
51
52     ## Ramp-up kit
53
54     - `docs/ALPHA_LAUNCH_READINESS.md`
55     -- `docs/data_pipeline/GOLDEN_VIDEOS.md`
56     -- `docs/R2_EVAL_METRIC_DESIGN.md`
57     -- `backend/eval/ablation.py`
58     -- `backend/eval/ablation_pdf.py`
59     -- `backend/eval/rally_seg_eval.py`
60     -- `tests/test_ablation.py`
61     -- `output/human_lovo_manifest.json`
62     +- `backend/eval/fusion_golden.py`
63     +- `backend/eval/fusion_audio.py`
64     +- `training/gen0/harness.py`
65     +- `eval_baselines/heuristic_lovo_n15.json`
66     +- `output/a7_remaining_default.out.log`
67     +- `output/a8_audio_augment.out.log`
68     +- `output/a6_r2_ablation.json`
69     +- `artifacts/rally_tal_gen0/0.1.1-n15-shadow.20260704/manifest.json`
70
71     ## Key decisions
72
73     -- R2/tolF1 is supported for the baseline-off to served milestone contrast on
74     -   the 15-video corpus: `Delta tolF1 = +0.225`, CI `[+0.165, +0.283]`,
75     -   sign `14/0`, `p = 0.000`.
76     -- This does not approve fusion/person/audio default changes. Current full-fusion
77     -   ablation still runs on only six videos because the newer nine do not yet have
78     -   full-fusion feature schema.
79     -- Branching discipline remains important: always fetch/pull first, branch from
80     -   `origin/master`, stage explicit paths, and verify `origin/master..HEAD` before
81     -   pushing.
82     +- A7: full-fusion schema and audio features are present locally for all 15 videos.
83     +- A8: person fusion and audio are suggestive but not statistically clean; do not
promote
84     +   either as a default. Gate A regresses under served windowing; keep it opt-
in/analysis-only.
85     +- R2/cap/inactivity remains the strongest quality result on n15: F1 `0.237 -> 0.472`,
86     +   tolF1 `0.138 -> 0.364`, merge rate `0.638 -> 0.135`; net over-seg guard passes.
87     +- A9: Gen-0 LogReg/TAL n15 shadow bundle beats the heuristic mean (`0.551 > 0.4456`)

```

```

88 + but is non-serving (`9/15` paired wins, `p=0.3036`; #172 and WASB C3 still block).
89 +- Branching discipline remains important: fetch/pull first, stage explicit paths, and
90 + verify `origin/master..HEAD` contains only this branch's commits before pushing.
91 diff --git a/backend/eval/fusion_audio.py b/backend/eval/fusion_audio.py
92 index b5c65ec4..7dff0c94 100644
93 --- a/backend/eval/fusion_audio.py
94 +++ b/backend/eval/fusion_audio.py
95 @@ -17,14 +17,17 @@
96     import json
97     import logging
98     import os
99 -from typing import Any, Dict, List
100 +from typing import Any, Dict, List, Optional, Sequence
101
102     from backend.eval import experiment
103     from backend.eval.fusion_compare import delta_stats
104     from backend.eval.fusion_golden import (
105         PERSON_FEATURE_NAMES,
106         SHUTTLE_FEATURE_NAMES,
107 +     SOURCE_MANIFEST_ENV,
108         _derive_video_path,
109 +     _filter_specs,
110 +     attach_source_video_paths,
111     )
112     from backend.eval.golden_manifest import load_resolved_manifest
113     from backend.pipeline.audio.hit_detector import detect_hits_in_video
114 @@ -33,9 +36,24 @@
115     logger = logging.getLogger(__name__)
116
117
118     -def augment(manifest_path: str, features_dir: str, k: float = 2.5) -> List[str]:
119     +def augment(
120     +     manifest_path: str,
121     +     features_dir: str,
122     +     k: float = 2.5,
123     +     source_manifest: Optional[str] = None,
124     +     only: Sequence[str] = (),
125     +) -> List[str]:
126         """Add the 3 audio features to each existing ``<name>_golden_features.json``
127         window."""
128         specs = load_resolved_manifest(manifest_path)
129         source_manifest = source_manifest or os.environ.get(SOURCE_MANIFEST_ENV)
130         if source_manifest:
131             updated = attach_source_video_paths(specs, source_manifest)
132             print(
133                 f"[fusion_audio] attached {updated}/{len(specs)} source video path(s) "
134                 f"from {source_manifest}",
135                 flush=True,
136             )
137             specs = _filter_specs(specs, only)
138             written: List[str] = []
139             for spec in specs:
140                 jp = os.path.join(features_dir, f"{spec['name']}_golden_features.json")
141                 @@ -45,6 +63,10 @@ def augment(manifest_path: str, features_dir: str, k: float = 2.5) ->
142                 List[str]:
143                     )
144                     continue
145                     video = _derive_video_path(spec)
146                     if not os.path.isfile(video):
147                         raise FileNotFoundError(
148                             f"{spec.get('name')}: source video not found for audio augmentation:
149 {video}"
150                     )
151                     print(
152                         f"[fusion_audio] {spec['name']}: detecting audio hits (k={k})...",

```

```

150         flush=True,
151     @@ -129,6 +151,21 @@ def main() -> None:
152         ap.add_argument("--manifest", default="output/human_lovo_manifest.json")
153         ap.add_argument("--features-dir", default="output")
154         ap.add_argument("--k", type=float, default=2.5, help="hit-detector sensitivity")
155     + ap.add_argument(
156     +         "--source-manifest",
157     +         default=None,
158     +         help=(
159     +             "collector golden_source_videos.json; overrides video paths for
source/proxy "
160     +             f"videos (or set {SOURCE_MANIFEST_ENV})"
161     +         ),
162     +     )
163     + ap.add_argument(
164     +         "--only",
165     +         action="append",
166     +         default=[],
167     +         metavar="VIDEO_ID",
168     +         help="augment one manifest video by name; repeatable",
169     +     )
170     + ap.add_argument(
171     +         "--augment",
172     +         action="store_true",
173     @@ -142,7 +179,13 @@ def main() -> None:
174         args = ap.parse_args()
175
176         if args.augment:
177             augment(args.manifest, args.features_dir, args.k)
178         + augment(
179         +         args.manifest,
180         +         args.features_dir,
181         +         args.k,
182         +         source_manifest=args.source_manifest,
183         +         only=args.only,
184         +     )
185         if args.compare:
186             videos = load_videos_with_audio(args.features_dir)
187             if len(videos) < 2:
188                 diff --git a/backend/eval/fusion_golden.py b/backend/eval/fusion_golden.py
189                 index aee6d7e5..0666a0ca 100644
190                 --- a/backend/eval/fusion_golden.py
191                 +++ b/backend/eval/fusion_golden.py
192             @@ -21,7 +21,7 @@
193                 import json
194                 import os
195                 import time
196                 -from typing import Any, Dict, List, Optional
197                 +from typing import Any, Dict, List, Optional, Sequence
198
199                 from backend.eval import distill_local
200                 from backend.eval.calibrate_wasb import load_golden_gts
201             @@ -36,6 +36,100 @@
202                 SHUTTLE_FEATURE_NAMES = list(distill_local.FEATURE_NAMES)
203                 # The 5 person-cue features (fusion_features.PERSON_FEATURE_NAMES).
204                 PERSON_FEATURE_NAMES = list(ff.PERSON_FEATURE_NAMES)
205                 +SOURCE_MANIFEST_ENV = "RALLY_GOLDEN_SOURCE_MANIFEST"
206                 +
207                 +
208                 +def is_full_fusion_feature_file(path: str) -> bool:
209                 +    """True when an existing golden feature JSON already has shuttle+person cue dicts.
210                 +
211                 +    Corpus-growth runs can leave leaner windowing-only JSONs with the same filename
pattern.
212                 +    Those are useful for gts/windowing analysis, but they are not the replayable fusion

```

```

213 +     substrate that ``fusion_compare`` and ``ablation`` need.
214 +     """
215 +
216 +     try:
217 +         with open(path, encoding="utf-8") as f:
218 +             data = json.load(f)
219 +     except (OSError, json.JSONDecodeError, TypeError):
220 +         return False
221 +     candidates = data.get("candidates")
222 +     if not isinstance(candidates, list):
223 +         return False
224 +     if not candidates:
225 +         return True
226 +     return all(
227 +         isinstance(candidate, dict)
228 +         and "shuttle" in candidate
229 +         and "person" in candidate
230 +         for candidate in candidates
231 +     )
232 +
233 +
234 +def load_source_video_paths(source_manifest_path: Optional[str]) -> Dict[str, str]:
235 +    """Load ``video_id -> local_path`` from the collector golden-source manifest.
236 +
237 +    The collector records source/proxy ownership and full-file MD5s. Only entries
marked
238 +    ``present_local`` with a concrete ``local_path`` can drive local GPU extraction.
239 +    """
240 +
241 +    if not source_manifest_path:
242 +        return {}
243 +    with open(source_manifest_path, encoding="utf-8") as f:
244 +        data = json.load(f)
245 +        entries = data.get("videos", data) if isinstance(data, dict) else data
246 +        if not isinstance(entries, list):
247 +            raise ValueError(f"source manifest must contain a videos list:
{source_manifest_path}")
248 +
249 +        paths: Dict[str, str] = {}
250 +        for entry in entries:
251 +            if not isinstance(entry, dict):
252 +                continue
253 +            video_id = entry.get("video_id") or entry.get("name")
254 +            local_path = entry.get("local_path")
255 +            if not video_id or not local_path:
256 +                continue
257 +            if entry.get("source_status", "present_local") != "present_local":
258 +                continue
259 +            paths[str(video_id)] = str(local_path)
260 +        return paths
261 +
262 +
263 +def attach_source_video_paths(
264 +    specs: Sequence[Dict[str, Any]], source_manifest_path: Optional[str]
265 +) -> int:
266 +    """Attach explicit ``video_path`` fields to manifest specs from the source
manifest.
267 +
268 +    Returns the number of specs updated. The collector source manifest is authoritative
269 +    for local source/proxy video location and overrides derived/stale manifest paths.
270 +    """
271 +
272 +    source_paths = load_source_video_paths(source_manifest_path)
273 +    updated = 0
274 +    for spec in specs:

```



```

275 +         name = str(spec.get("name", ""))
276 +         source_path = source_paths.get(name)
277 +         if not source_path:
278 +             continue
279 +         if spec.get("video_path") == source_path:
280 +             continue
281 +         spec["video_path"] = source_path
282 +         updated += 1
283 +     return updated
284 +
285 +
286 +def _filter_specs(
287 +    specs: Sequence[Dict[str, Any]], only: Sequence[str]
288 +) -> List[Dict[str, Any]]:
289 +    if not only:
290 +        return list(specs)
291 +    wanted = list(dict.fromkeys(only))
292 +    wanted_set = set(wanted)
293 +    filtered = [spec for spec in specs if str(spec.get("name")) in wanted_set]
294 +    found = {str(spec.get("name")) for spec in filtered}
295 +    missing = [name for name in wanted if name not in found]
296 +    if missing:
297 +        raise ValueError(f"--only requested unknown video(s): {' '.join(missing)}")
298 +    return filtered
299
300
301     def _derive_video_path(spec: Dict[str, Any]) -> str:
302 @@ -58,6 +152,9 @@ def extract_video(
303         iou: float = 0.5,
304         log_every_sec: float = 15.0,
305         telemetry: Optional[Any] = None,
306 +    use_gpu: bool = True,
307 +    detector_model: Optional[str] = None,
308 +    max_gpu_temp_c: Optional[float] = None,
309     ) -> Dict[str, Any]:
310         """One golden video -> a feature record (shuttle + person features per candidate
window).
311
312     @@ -77,11 +174,18 @@ def extract_video(
313         # (parity with the shuttle net-crossing gate) rather than hard-coding an
orientation.
314         axis, net_px, _span = rally_gate.estimate_play_axis(points)
315         video_path = _derive_video_path(spec)
316 +    if not os.path.isfile(video_path):
317 +        raise FileNotFoundError(
318 +            f"{spec.get('name')}: source video not found for fusion extraction:
{video_path}"
319 +        )
320
321         if detector is None:
322             from backend.pipeline.detectors.person_detector import PersonDetector
323
324 -            detector = PersonDetector({"indexing": {"use_gpu": True}})
325 +            indexing_cfg: Dict[str, Any] = {"use_gpu": use_gpu}
326 +            if detector_model:
327 +                indexing_cfg["detector_model"] = detector_model
328 +            detector = PersonDetector({"indexing": indexing_cfg})
329
330         # Single-pass person detection over ALL candidate windows (open the video ONCE,
read
331         # forward ? no per-window seek). Then compute features per window (pure, no GPU).
332 @@ -95,8 +199,17 @@ def extract_video(
333         state = {"last": t0}
334
335         def _progress(done: int, total: int) -> None:

```

```

336 +         rec = None
337 +         if telemetry is not None:
338 -             telemetry.snapshot(done, total, phase=spec["name"])
339 +             rec = telemetry.snapshot(done, total, phase=spec["name"])
340 +         if max_gpu_temp_c is not None and rec is not None:
341 +             gpu = rec.get("gpu") or {}
342 +             temp = gpu.get("temp_c")
343 +             if temp is not None and float(temp) >= max_gpu_temp_c:
344 +                 raise RuntimeError(
345 +                     f"{spec['name']}: GPU temperature {float(temp):.1f}C reached "
346 +                     f"--max-gpu-temp-c {max_gpu_temp_c:.1f}; aborting before thermal
shutdown"
347 +                 )
348 +             now = time.monotonic()
349 +             if now - state["last"] >= log_every_sec or done >= total:
350 +                 el = now - t0
351 @@ -160,8 +273,23 @@ def run(
352     iou: float = 0.5,
353     fresh: bool = False,
354     smallest_first: bool = False,
355 +     source_manifest: Optional[str] = None,
356 +     only: Sequence[str] = (),
357 +     missing_fusion_only: bool = False,
358 +     use_gpu: bool = True,
359 +     detector_model: Optional[str] = None,
360 +     max_gpu_temp_c: Optional[float] = None,
361 ) -> List[str]:
362     specs = load_resolved_manifest(manifest_path)
363 +     source_manifest = source_manifest or os.environ.get(SOURCE_MANIFEST_ENV)
364 +     if source_manifest:
365 +         updated = attach_source_video_paths(specs, source_manifest)
366 +         print(
367 +             f"[fusion_golden] attached {updated}/{len(specs)} source video path(s) "
368 +             f"from {source_manifest}",
369 +             flush=True,
370 +         )
371 +     specs = _filter_specs(specs, only)
372     if smallest_first:
373         # Process small videos first for quick feedback (the slow full match lands
last);
374         # the LOVO result is order-independent. Proxy size by the trajectory CSV (one
row/frame).
375 @@ -187,9 +315,19 @@ def run(
376     written: List[str] = []
377     for vi, spec in enumerate(specs, 1):
378         out = os.path.join(out_dir, f"{spec['name']}_golden_features.json")
379 -         # Resumable: skip a video whose JSON already exists (a re-run picks up where a
prior
380 -         # run left off instead of re-detecting). --fresh forces a redo.
381 -         if not fresh and os.path.isfile(out):
382 +             existing = os.path.isfile(out)
383 +             existing_full = existing and is_full_fusion_feature_file(out)
384 +             # Resumable: skip a video whose full-fusion JSON already exists. A leaner
385 +             # windowing-only JSON is stale for A7, so re-extract it even without --fresh.
386 +             if missing_fusion_only and not fresh and existing_full:
387 +                 print(
388 +                     f"[fusion_golden] ({vi}/{len(specs)}) {spec['name']}: full-fusion "
389 +                     f"features already present -> {out}",
390 +                     flush=True,
391 +                 )
392 +                 written.append(out)
393 +                 continue
394 +             if not fresh and existing and existing_full:
395 +                 print(
396 +                     f"[fusion_golden] ({vi}/{len(specs)}) {spec['name']}: already extracted

```

```

"
397         f"-> {out} (use --fresh to redo)",
398     @@ -197,6 +335,12 @@ def run(
399         )
400         written.append(out)
401         continue
402     +     if not fresh and existing and not existing_full:
403     +         print(
404     +             f"[fusion_golden] ({vi}/{len(specs)}) {spec['name']}: existing JSON "
405     +             "is not full-fusion schema; re-extracting...",
406     +             flush=True,
407     +         )
408         print(
409             f"[fusion_golden] ({vi}/{len(specs)}) {spec['name']}: extracting...",
410             flush=True,
411     @@ -205,7 +349,15 @@ def run(
412         if telem is not None:
413             telem.event("video_start", spec["name"])
414         try:
415     -         data = extract_video(spec, frame_skip=frame_skip, iou=iou, telemetry=telem)
416     +         data = extract_video(
417     +             spec,
418     +             frame_skip=frame_skip,
419     +             iou=iou,
420     +             telemetry=telem,
421     +             use_gpu=use_gpu,
422     +             detector_model=detector_model,
423     +             max_gpu_temp_c=max_gpu_temp_c,
424     +         )
425         except Exception as e: # noqa: BLE001 - record machine state at death, then
re-raise
426             if telem is not None:
427                 telem.event("error", spec["name"], err=str(e))
428     @@ -246,6 +398,47 @@ def main() -> None:
429         action="store_true",
430         help="process smaller videos first for quick feedback (LOVO is order-
independent)",
431     )
432     +     ap.add_argument(
433     +         "--source-manifest",
434     +         default=None,
435     +         help=(
436     +             "collector golden_source_videos.json; overrides video paths for
source/proxy "
437     +             f"videos (or set {SOURCE_MANIFEST_ENV})"
438     +         ),
439     +     )
440     +     ap.add_argument(
441     +         "--only",
442     +         action="append",
443     +         default=[],
444     +         metavar="VIDEO_ID",
445     +         help="process one manifest video by name; repeatable",
446     +     )
447     +     ap.add_argument(
448     +         "--missing-fusion-only",
449     +         action="store_true",
450     +         help="skip videos whose existing JSON already has full shuttle+person fusion
schema",
451     +     )
452     +     ap.add_argument(
453     +         "--cpu",
454     +         action="store_true",
455     +         help="run torchvision person detection on CPU instead of CUDA",
456     +     )

```

```

457 +     ap.add_argument(
458 +         "--detector-model",
459 +         default=None,
460 +         choices=[
461 +             "fasterrcnn_resnet50_fpn",
462 +             "fasterrcnn_mobilenet_v3_large_fpn",
463 +             "retinanet_resnet50_fpn",
464 +         ],
465 +         help="torchvision detector model; default keeps the configured Faster R-CNN
ResNet",
466 +     )
467 +     ap.add_argument(
468 +         "--max-gpu-temp-c",
469 +         type=float,
470 +         default=None,
471 +         help="abort extraction if telemetry sees GPU temperature at or above this
value",
472 +     )
473     args = ap.parse_args()
474     run(
475         args.manifest,
476         @@ -254,6 +447,12 @@ def main() -> None:
477         args.iou,
478         fresh=args.fresh,
479         smallest_first=args.smallest_first,
480 +         source_manifest=args.source_manifest,
481 +         only=args.only,
482 +         missing_fusion_only=args.missing_fusion_only,
483 +         use_gpu=not args.cpu,
484 +         detector_model=args.detector_model,
485 +         max_gpu_temp_c=args.max_gpu_temp_c,
486     )
487
488
489     diff --git a/docs/ALPHA_LAUNCH_READINESS.md b/docs/ALPHA_LAUNCH_READINESS.md
490     index 5f550386..ea59c09b 100644
491     --- a/docs/ALPHA_LAUNCH_READINESS.md
492     +++ b/docs/ALPHA_LAUNCH_READINESS.md
493     @@ -1,6 +1,6 @@
494     # Alpha Launch Readiness Plan
495
496     -> **Status:** `DRAFT` - cross-repo execution tracker; **Owner:** @avidullu;
**Created:** 2026-07-03; **Last updated:** 2026-07-03
497     +> **Status:** `DRAFT` - cross-repo execution tracker; **Owner:** @avidullu;
**Created:** 2026-07-03; **Last updated:** 2026-07-04
498     > **Lifecycle:** `DRAFT` -> IN PROGRESS -> DONE -> archived`
499     > **Tracking anchors:** Section 7 is the progress tracker and source of truth; indexed
in `docs/README.md` and `docs/DOC_STATUS.md`.
500     > **Relation to existing docs:** consolidates the launch-relevant pending items from
`NEXT_STEPS.md`, `ROADMAP.md`, `docs/data_pipeline/GOLDEN_VIDEOS.md`,
`R2_EVAL_METRIC_DESIGN.md`, `COMPUTE_DECOUPLED_SERVING/`, and the sibling `kheIsutra` Studio
docs.
501     @@ -10,7 +10,7 @@
502
503     ## 0. TL;DR
504
505     -The corpus is now large enough to move from "grow labels first" into alpha-readiness
work, but the 15-video headline is not itself a launch gate pass. `[verified]` The golden
manifest has 15 badminton videos; `[analysis]` the local manifest needed path rebasing before
manifest-driven commands ran; `[analysis]` only 6 of 15 feature JSONs had the full fusion
schema; `[analysis]` the n=15 quality baseline is stale and must be intentionally refreshed
before it can gate anything.
506     +The corpus is now large enough to move from "grow labels first" into alpha-readiness
work, but the 15-video headline is not itself a launch gate pass. `[verified]` The golden
manifest has 15 badminton videos; `[analysis]` A2-A7 made the local manifest/path and full-

```

fusion/audio feature surface reproducible on all 15; `[analysis]` A4/A5 refreshed the n=15 nightly and heuristic floors; `[analysis]` A8/A9 still reject served Gate A default promotion and owned-model serving claims on the current evidence.

507

508 The first controlled alpha should be a hosted, authenticated product loop: upload a video, process it, select rallies, compile, and download a reel. The owned-model/commercial rally-detection claim remains a separate NO-GO until the ship-gate target is ratified and the WASB checkpoint provenance is resolved.

509

510 @@ -31,8 +31,9 @@ This doc separates those into executable workstreams. The goal is to reach an in

511 | D1 | **Product alpha and owned-model launch are separate decisions.** |
`NEXT\_STEPS.md` NO-GO banner, 2026-06-30; verified 2026-07-03 | We can launch a controlled hosted product loop without claiming the owned model is commercially cleared. |

512 | D2 | **First alpha must be page-driven, not SSH/CLI-driven.** | `NEXT\_STEPS.md` P0 UI-driven/shareable item; verified 2026-07-03 | Hosted service, edge auth, upload, job polling, compile, and download are launch-critical. |

513 | D3 | **The 15-video corpus is now a measurement asset, not a free launch pass.** |
`[verified]` `docs/data\_pipeline/GOLDEN\_VIDEOS.md`; `[analysis]` local eval runs, 2026-07-03 | Rebase/portability, feature completeness, baseline refresh, and gate ratification come before quality claims. |

514 -| D4 | **Do not promote served Gate A by default from the current n=15 evidence.** |
`[analysis]` `backend.eval.serve\_contrast` run, 2026-07-03 | Proposal-side lift does not survive served-windowing safety; keep it as an opt-in/analysis lever until owner ratification. |

515 +| D4 | **Do not promote served Gate A by default from the current n=15 evidence.** |
`[analysis]` `backend.eval.serve\_contrast` run, 2026-07-04 | Proposal-side lift does not survive served-windowing safety; keep it as an opt-in/analysis lever until owner ratification. |

516 | D5 | **P10/P11 corpus promotion and train/re-eval are alpha-plus unless owner reopens D13.** | sibling `khelsutra/docs/STUDIO\_MEDIA\_LIFECYCLE.md`; verified 2026-07-03 | First alpha can use manual/offline corpus promotion; automated flywheel follows after serving and gates are stable. |

517 +| D6 | **Gen-0 n=15 shadow training is non-serving.** | `[analysis]`
`training.gen0.harness` run, 2026-07-04 | Learned mean beats the heuristic floor, but the paired sign test is not significant; #172 and WASB C3 remain hard blockers. |

518

519 ## 3. Verified snapshot, 2026-07-03

520

521 @@ -50,10 +51,11 @@ Corpus/eval facts:

522

523 - `[verified]` `docs/data\_pipeline/GOLDEN\_VIDEOS.md` says
`output/human\_lovo\_manifest.json` is the ingested golden corpus at `n = 15`.

524 - `[analysis]` The local manifest had stale absolute paths. A temporary rebased manifest found all 15 trajectories and labels in this workspace.

525 -- `[analysis]` There are 15 `output/\*\_golden\_features.json` files, but only 6 carry full shuttle/person/audio candidate features. The newer 9 are windowing-only schema and are skipped by fusion/person eval.

526 -- `[analysis]` `python -m backend.eval.fusion\_compare --features-dir output` loaded only 6 videos and measured person-fusion `Delta F1 = -0.021`; not promotable.

527 -- `[analysis]` `python -m backend.eval.serve\_contrast --manifest <rebased-n15>` measured proposal Gate A mean `Delta F1 +0.0287`, but served mean `Delta F1 -0.0328`; not served-safe.

528 -- `[analysis]` `python -m training.gen0.harness --manifest <rebased-n15> --floor eval\_baselines/heuristic\_lovo\_n6.json --version analysis-n15 --out <temp-out>` produced learned LOVO mean F1 `0.551`, `ship=False`; this pre-A5 run used the stale n=6 floor, so future comparisons should use `eval\_baselines/heuristic\_lovo\_n15.json`.

529 +- `[analysis]` A7 generated full shuttle/person fusion schema for all 15 videos and A8 audio augmentation populated audio features for all 15.

530 +- `[analysis]` `python -m backend.eval.fusion\_compare --features-dir output` now runs over all 15 and measures person-fusion `Delta F1 = +0.008`; the paired CI crosses zero, so person fusion is not promotable by itself.

531 +- `[analysis]` `python -m backend.eval.fusion\_audio --features-dir output --compare` measures audio `Delta F1 = +0.018`; the paired CI crosses zero, so audio is a shadow/suggestive cue, not a default-flip.

532 +- `[analysis]` `python -m backend.eval.serve\_contrast --manifest output/human\_lovo\_manifest.json` measured proposal Gate A mean `Delta F1 +0.0287`, but served

```

mean `Delta F1 -0.0328`; not served-safe.
533   +- `[analysis]` `python -m training.gen0.harness --manifest
output\human_lovo_manifest.json --floor eval_baselines\heuristic_lovo_n15.json --version
0.1.1-n15-shadow.20260704 --out artifacts\rally_tal_gen0` produced learned LOVO mean F1 `0.551`,
floor passed versus heuristic `0.446`, but sign test `9/15`, `p = 0.3036`, `ship=False`.
534   - `[analysis]` `python scripts/nightly_regression.py --manifest <rebased-n15>
--features-dir output --no-report` reported default/milestone tolF1 `0.3636` vs baseline-off
tolF1 `0.1382`, but marked the frozen baseline stale due to config/corpus changes.
535   - `[analysis]` Before A6, `python -m backend.eval.ablation --features-dir output
--granularity group` failed at import due to a circular import between
`backend/eval/ablation.py` and `backend/eval/ablation_pdf.py`; A6 fixes the module entrypoint
and keeps full-fusion ablation scoped to the 6 videos with full schema until A7.
536
537   @@ -97,13 +99,13 @@ Legend: TODO = not started, IN PROGRESS = actively being worked,
DONE = shipped/
538   | A0 | Create this alpha readiness tracker and index it in docs | `badminton-highlight-
indexer` | - | No | DONE | #451 |
539   | A1 | Stand up alpha serving endpoint: Cloud Run/GPU or approved host, health check,
edge auth ON, upload -> process -> jobs -> compile -> download verified | `khealsutra` +
`badminton-highlight-indexer` | A0 | Owner spend / CF config | TODO | - |
540   | A2 | Make the n=15 golden manifest portable/reproducible; add a path/corpus smoke
check that fails loudly when labels or trajectories are missing | `badminton-highlight-indexer`
| A0 | No | DONE | #452 |
541   -| A3 | Promote the 15-video source-video/MD5 records into the collector/vault path;
reconcile collector's six-video source manifest with the 15-video eval corpus | `sports-data-
collector` + vault | A2 | Owner upload/storage scope | TODO | - |
542   +| A3 | Promote the 15-video source-video/MD5 records into the collector/vault path;
reconcile collector's six-video source manifest with the 15-video eval corpus | `sports-data-
collector` + vault | A2 | Owner upload/storage scope | IN PROGRESS | collector #62 |
543   | A4 | Refresh the n=15 nightly regression baseline intentionally and record the exact
command/output; keep stale-baseline warnings until reviewed | `badminton-highlight-indexer` | A2
| Reviewer sign-off | DONE | #453 |
544   | A5 | Recompute the heuristic n=15 floor and replace the stale `heuristic_lovo_n6`
floor for future Gen-0 comparisons | `badminton-highlight-indexer` | A2 | No | DONE | #454 |
545   -| A6 | Fix `backend.eval.ablation` CLI circular import, then run the R2/tolF1 ablation
on the n=15 corpus | `badminton-highlight-indexer` | A2 | No | IN PROGRESS | this PR |
546   -| A7 | Regenerate or upconvert full-fusion feature JSONs for the newer 9 videos so
fusion/person/audio analyses run over all 15 | `badminton-highlight-indexer` | A2 | GPU/data
availability | TODO | - |
547   -| A8 | Re-run `fusion_compare`, group ablation, and served contrast after A7;
explicitly promote/reject person fusion, Gate A served-default, and any other R-series default
flips | `badminton-highlight-indexer` | A6, A7 | Reviewer sign-off | TODO | - |
548   -| A9 | Run Gen-0/TemporalMaxer shadow training against the refreshed n=15 floor;
produce a non-serving artifact bundle and gate verdict | `badminton-highlight-indexer` | A5, A8
| Owner approval for M0.4 scope | TODO | - |
549   +| A6 | Fix `backend.eval.ablation` CLI circular import, then run the R2/tolF1 ablation
on the n=15 corpus | `badminton-highlight-indexer` | A2 | No | DONE | #455 |
550   +| A7 | Regenerate or upconvert full-fusion feature JSONs for the newer 9 videos so
fusion/person/audio analyses run over all 15 | `badminton-highlight-indexer` | A2 | GPU/data
availability | IN PROGRESS | this PR |
551   +| A8 | Re-run `fusion_compare`, group ablation, and served contrast after A7;
explicitly promote/reject person fusion, Gate A served-default, and any other R-series default
flips | `badminton-highlight-indexer` | A6, A7 | Reviewer sign-off | IN PROGRESS | this PR |
552   +| A9 | Run Gen-0/TemporalMaxer shadow training against the refreshed n=15 floor;
produce a non-serving artifact bundle and gate verdict | `badminton-highlight-indexer` | A5, A8
| Owner approval for M0.4 scope | IN PROGRESS | this PR |
553   | A10 | Decide and record the alpha ship-gate target: metric, corpus slice, minimum
threshold, significance rule, and what claims are allowed | Owner + `badminton-highlight-
indexer` | A4, A5, A8 | Owner decision | GATED | - |
554   | A11 | Resolve or explicitly defer WASB C3 provenance for alpha messaging; if
deferred, document "no commercial model sharing" copy guardrail | Owner/legal | A10 |
Legal/provenance | GATED | - |
555   | A12 | Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector
promotion, `train`/`reeval` job, delta-F1 surface, and alignment with
`COMPUTE_DECOUPLED_SERVING` M11 | `khealsutra` + `badminton-highlight-indexer` + collector | A1,

```

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A3 | Owner D13 reopen (`khelsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`) | GATED | - |
556     @@ -174,22 +176,57 @@ python -m backend.eval.rally_seg_eval --manifest
output\human_lovo_manifest.json
557
558     Result: the ablation CLI now runs, but full-fusion feature ablation still reports `n=6`
because only the original six feature JSONs carry the full fusion schema; A7 must
regenerate/upconvert the newer nine before fusion/person/audio promotion decisions. The n=15
R2/tolF1 contrast is positive for the served milestone versus baseline-off: baseline/off `tolF1
= 0.138`, served `tolF1 = 0.364`, `Delta tolF1 = +0.225` with CI `[+0.165, +0.283]`, sign
`14/0`, `p = 0.000`; `Delta F1@0.5 = +0.236` with CI `[+0.161, +0.313]`; merge rate improves
from `0.638` to `0.135`. Net over-segmentation guard passes (`1.276 -> 0.378`), while the strict
segment-count rule remains blocked by increased splits, so R2/tolF1 is the supported gate for
the baseline-off to served milestone contrast rather than a blanket fusion/default-flip
approval.
559
560     +A7 full-fusion/audio feature refresh record:
561     +
562     +```powershell
563     +python -m backend.eval.fusion_golden --manifest output\human_lovo_manifest.json --out-
dir output --source-manifest ..\sports-data-collector\data\golden_source_videos.json --missing-
fusion-only --smallest-first --max-gpu-temp-c 87
564     +python -m backend.eval.fusion_audio --manifest output\human_lovo_manifest.json
--features-dir output --source-manifest ..\sports-data-collector\data\golden_source_videos.json
--augment
565     +```
566     +
567     +Result: the GPU pass completed without a thermal abort after fans were set to max
performance. Full-fusion schema is now present for `15/15` feature JSONs, and audio features are
present for `15/15`. The large regenerated clips completed as follows: `mahadevpura_1` `1`
window / `0` positives / `10` GT, `GX020094` `99` / `30` / `40`, `GX030094` `79` / `21` / `41`,
`GX010141` `26` / `11` / `29`, `GX010137` `56` / `31` / `34`, `Boxhill_Doubles` `59` / `12` /
`43`, `mahadevpura_2` `14` / `2` / `40`, `GX010128` `51` / `15` / `52`, `Badminton_BXH_2` `29` /
`1` / `44`. The source manifest supplied all 15 source-video paths, including the F-drive
Boxhill originals.
568     +
569     +A8 fusion/default-decision record:
570     +
571     +```powershell
572     +python -m backend.eval.fusion_compare --features-dir output
573     +python -m backend.eval.fusion_audio --features-dir output --compare
574     +python -m backend.eval.ablation --features-dir output --granularity group
575     +python -m backend.eval.serve_contrast --manifest output\human_lovo_manifest.json
576     +python -m backend.eval.rally_seg_eval --manifest output\human_lovo_manifest.json
--features-dir output --preset served --max-window-duration 0 --inactivity-end 0 --vs served
--vs-max-window-duration 6 --vs-inactivity-end 1.5 --json-out output\A6_r2_ablation.json
577     +```
578     +
579     +Result:
580     +
581     +- Person fusion is **not promoted**: shuttle-only F1 `0.302`, shuttle+person F1
`0.311`, `Delta F1 +0.008`, paired CI `[-0.040, +0.055]`, sign `8W/6L`, `p = 0.791`.
582     +- Audio is **shadow/suggestive, not promoted**: shuttle+person F1 `0.311`,
shuttle+person+audio F1 `0.329`, `Delta F1 +0.018`, paired CI `[-0.017, +0.053]`, sign `10W/4L`,
`p = 0.180`.
583     +- Group ablation remains non-significant for cue promotion: audio `+0.018` CI `[-0.017,
+0.053]`; shuttle `+0.013` CI `[-0.002, +0.029]`; gate_a `-0.001` CI `[-0.012, +0.008]`; person
`-0.009` CI `[-0.045, +0.021]`.
584     +- Gate A served default is **rejected**: proposal mean `Delta F1 +0.0287`, CI
`[-0.0222, +0.0680]`; served mean `Delta F1 -0.0328`, CI `[-0.0799, +0.0052]`, `regresses=True`.
585     +- R2/cap/inactivity remains the strong quality result: served candidate F1 `0.472` vs
baseline-off `0.237`, `Delta F1 +0.236`, CI `[+0.161, +0.313]`; tolF1 `0.364` vs `0.138`, `Delta
tolF1 +0.225`, CI `[+0.165, +0.283]`; merge rate `0.135` vs `0.638`. Net over-seg guard passes,
strict per-video split-count guard still blocks blanket promotion language.
586     +
587     +A9 Gen-0 shadow-training record:

```

```

588 +
589 +```powershell
590 +python -m training.gen0.harness --manifest output\human_lovo_manifest.json --floor
eval_baselines\heuristic_lovo_n15.json --out artifacts\rally_tal_gen0 --version
0.1.1-n15-shadow.20260704
591 +```
592 +
593 +Result: current Gen-0 LogReg/TAL harness produced local bundle
`artifacts\rally_tal_gen0\0.1.1-n15-shadow.20260704` with learned LOVO mean F1 `0.551` versus
heuristic floor `0.4456`, so `floor_passed=True`. The gate remains **non-serving**: paired sign
test `9/15`, `p = 0.3036`, `ship=False`; blockers are uncalibrated ship-gate target #172 and
WASB checkpoint provenance C3. TemporalMaxer itself is not implemented in this branch; the
evidence supports either closing A9 as a Gen-0 shadow baseline or opening a follow-up M0.4
scorer-swap PR.
594 +
595 ## 9. Open questions
596
597 1. **Owner:** Is the first alpha allowed to use manual/offline corpus promotion, or
should P10/P11 be reopened before any tester wave?
598 2. **Owner/platform:** Which hosted compute target is the next stand-up: Cloud Run GPU,
GCE L4 VM, or another approved GPU host?
599 3. **Owner/legal:** What is the acceptable alpha wording while WASB C3 remains
unresolved?
600 4. **Engineering:** Should the portable corpus manifest live as a generated artifact, a
checked-in redacted manifest, or both?
601 -5. **Engineering:** Should n=15 feature regeneration target full fusion schema for
every clip, or should the eval loaders support mixed schemas with explicit capability tags?
602 +5. **Engineering:** Should A9 close on the current Gen-0 non-serving bundle, or should
the actual TemporalMaxer scorer swap be implemented before marking A9 done?
603
604 ## 10. Definition of done
605
606 - [ ] Invited tester can complete the hosted record/upload -> reel download CUJ without
SSH.
607 - [ ] Alpha service is gated behind the chosen auth posture and has a documented
disable/rollback path.
608 -- [ ] n=15 corpus is portable enough that manifest-driven quality commands do not rely
on stale local absolute paths.
609 -- [ ] n=15 nightly baseline and heuristic floor are refreshed with recorded commands.
610 -- [ ] R2/tolF1 gate is promoted or rejected with a recorded ablation result.
611 -- [ ] Fusion/person/Gate-A served-default decisions are explicitly promoted or rejected
on the 15-video corpus.
612 +- [x] n=15 corpus is portable enough that manifest-driven quality commands do not rely
on stale local absolute paths.
613 +- [x] n=15 nightly baseline and heuristic floor are refreshed with recorded commands.
614 +- [x] R2/tolF1 gate is promoted or rejected with a recorded ablation result.
615 +- [x] Fusion/person/Gate-A served-default decisions are explicitly promoted or rejected
on the 15-video corpus.
616 - [ ] Ship-gate target and allowed alpha claims are recorded.
617 - [ ] WASB C3 status is recorded and reflected in alpha messaging.
618 - [ ] P10/P11 are either reopened with a committed sequence or explicitly deferred for
alpha-plus.
619 @@ -226,3 +263,4 @@ python -m backend.eval.rally_seg_eval --manifest
output\human_lovo_manifest.json
620 ### Changelog
621
622 - 2026-07-03 - Created from a synced five-repo review of `NEXT_STEPS`, `ROADMAP`,
golden-corpus docs, local eval artifacts, and the current 15-video corpus state.
623 +- 2026-07-04 - Recorded A7 full-fusion/audio regeneration, A8 n15 fusion/default
decisions, and A9 Gen-0 n15 shadow-training gate verdict.
624 diff --git a/tests/test_ablation.py b/tests/test_ablation.py
625 index a4491d7c..b5450f7b 100644
626 --- a/tests/test_ablation.py
627 +++ b/tests/test_ablation.py
628 @@ -10,9 +10,10 @@

```



```

629         - the honest-stats contract ? every row ships a CI + sign test, and no verdict is
stronger
630         than the stats support (the no-over-claim contract).
631         - a valid non-trivial PDF is produced (asserted via pypdf page count > 0).
632         - the golden litmus reproduction ? on the 6 real ``output/*_golden_features.json``
the runner
633         - reproduces the hand-measured Gate-A result (?F1 ? +0.074, 6/6 wins, p ? 0.031).
Skips
634         - cleanly when the golden JSONs are absent (CI), runs when present (owner box).
635         + - the golden litmus reproduction ? on real ``output/*_golden_features.json`` files
the runner
636         + reproduces the hand-measured Gate-A result for either the legacy n=6 substrate or
the refreshed
637         + n=15 A7 substrate. Skips cleanly when the golden JSONs are absent (CI), runs when
present
638         + (owner box).
639         ""
640
641         from __future__ import annotations
642         @@ -517,7 +518,7 @@ def test_load_videos_skips_windowing_only_schema(tmp_path):
643
644
645         def _golden_present() -> bool:
646         -     """True iff ?6 FULL-FUSION-SCHEMA golden files exist (the litmus needs the
shuttle/person cue
647         +     """True iff >=6 FULL-FUSION-SCHEMA golden files exist (the litmus needs the
shuttle/person cue
648             columns). Counts only ablatabe files ? leaner velocity-windowing files are skipped
by
649             ``load_videos``, so counting them would trip the litmus on an un-ablatable corpus
(the n=11
650             mixed-schema bug)."""
651         @@ -540,29 +541,42 @@ def _golden_present() -> bool:
652             reason="golden output/*_golden_features.json absent (run fusion_golden first)",
653         )
654         def test_litmus_reproduces_gate_a():
655         -     """LITMUS: on the 6 real golden feature JSONs the runner reproduces the hand-
measured Gate-A
656         -     result ? marginal ?F1 ? +0.074, 6/6 wins, p ? 0.031, CI ? [+0.042, +0.104], within
tolerance.
657         -     This validates the whole framework against the recorded ground truth. Skips cleanly
in CI
658         -     where the golden substrate is absent; runs on the owner box where it is present."""
659         +     """LITMUS: on real golden feature JSONs the runner reproduces the recorded Gate-A
result.
660         +
661         +     The legacy n=6 owner-box substrate preserves the 2026-06-14 hand measurement. After
A7, the
662         +     refreshed owner-box substrate has n=15 full-fusion files; that result is
intentionally weaker
663         +     and supports "keep structural, do not promote served default".
664         +     ""
665             videos = ablation.load_videos(GOLDEN_DIR)
666             assert len(videos) == 6
667         +     assert len(videos) in {6, 15}
668             cfg = ablation.litmus_config()
669             report = run_ablation(videos, cfg)
670             assert len(report.rows) == 1
671             row = report.rows[0]
672             assert row.signal == "gate_a"
673         -     # The recorded QUALITY_ITERATIONS truth (2026-06-14), reproduced within tolerance.
674         -     assert row.delta_f1 == pytest.approx(0.074, abs=0.005)
675         -     assert row.sign_wins == 6 and row.sign_losses == 0
676         -     assert row.sign_p_value == pytest.approx(0.031, abs=0.005)
677         -     assert row.ci_low == pytest.approx(0.042, abs=0.01)

```

```

678 - assert row.ci_high == pytest.approx(0.104, abs=0.01)
679 - assert row.ci_excludes_zero is True
680 + if len(videos) == 6:
681 +     # The recorded QUALITY_ITERATIONS truth (2026-06-14), reproduced within
tolerance.
682 +     assert row.delta_f1 == pytest.approx(0.074, abs=0.005)
683 +     assert row.sign_wins == 6 and row.sign_losses == 0
684 +     assert row.sign_p_value == pytest.approx(0.031, abs=0.005)
685 +     assert row.ci_low == pytest.approx(0.042, abs=0.01)
686 +     assert row.ci_high == pytest.approx(0.104, abs=0.01)
687 +     assert row.ci_excludes_zero is True
688 +     # C4: over-segmentation drops 1.68 -> 1.01 when Gate-A is on (full).
689 +     assert row.seg_ratio_full == pytest.approx(1.01, abs=0.05)
690 +     assert row.seg_ratio_ablated == pytest.approx(1.68, abs=0.05)
691 + else:
692 +     # The A7 refreshed n=15 substrate: proposal-side structural signal, but CI
crosses zero.
693 +     assert row.delta_f1 == pytest.approx(0.0287, abs=0.005)
694 +     assert row.sign_wins == 12 and row.sign_losses == 1
695 +     assert row.sign_p_value == pytest.approx(0.0034, abs=0.002)
696 +     assert row.ci_low == pytest.approx(-0.020, abs=0.01)
697 +     assert row.ci_high == pytest.approx(0.066, abs=0.01)
698 +     assert row.ci_excludes_zero is False
699 +     assert row.seg_ratio_full == pytest.approx(0.79, abs=0.05)
700 +     assert row.seg_ratio_ablated == pytest.approx(1.36, abs=0.05)
701     # Gate-A is structural ? reported but never auto-promoted to earns_keep.
702     assert row.verdict == ablation.STRUCTURAL_PROTECTED
703 - # C4: over-segmentation drops 1.68 ? 1.01 when Gate-A is on (full).
704 - assert row.seg_ratio_full == pytest.approx(1.01, abs=0.05)
705 - assert row.seg_ratio_ablated == pytest.approx(1.68, abs=0.05)
706
707
708 @pytest.mark.skipif(
709 @@ -572,7 +586,7 @@ def test_litmus_report_is_well_formed():
710     """The litmus report is a proper AblationReport (provenance + a sorted single
row)."""
711     report = run_ablation(ablation.load_videos(GOLDEN_DIR), ablation.litmus_config())
712     assert isinstance(report, AblationReport)
713 - assert report.num_videos == 6
714 + assert report.num_videos in {6, 15}
715     assert report.full_set_signals == ["gate_a"]
716     assert report.label_set_hash
717     out = ablation.format_table(report)
718     diff --git a/tests/test_audio_features.py b/tests/test_audio_features.py
719     index 26028591..d962b5e4 100644
720     --- a/tests/test_audio_features.py
721     +++ b/tests/test_audio_features.py
722     @@ -118,6 +118,7 @@ def
test_augment_rebases_stale_manifest_before_deriving_video_path(tmp_path, mon
723         out.mkdir(parents=True)
724         traj.write_text("x\n", encoding="utf-8")
725         labels.write_text("start_time,end_time\n0,1\n", encoding="utf-8")
726 +     labels.with_name("Foo.MP4").write_text("video\n", encoding="utf-8")
727     manifest = out / "human_lovo_manifest.json"
728     manifest.write_text(
729         json.dumps(
730     @@ -163,6 +164,80 @@ def _fake_detect(video_path, k=2.5):
731         assert data["candidates"][0]["audio"]["audio_hit_count"] == 1.0
732
733
734 +def test_augment_uses_source_manifest_video_path(tmp_path, monkeypatch):
735 +    workspace = tmp_path / "workspace"
736 +    repo = workspace / "badminton-highlight-indexer"
737 +    out = repo / "output"
738 +    traj = workspace / "Annotation Setup" / "Trajectories" / "foo.csv"

```

```

739 + labels = workspace / "Annotation Setup" / "Golden Labelled" / "Foo.rallies.csv"
740 + source_video = workspace / "source" / "Foo_proxy.mp4"
741 + traj.parent.mkdir(parents=True)
742 + labels.parent.mkdir(parents=True)
743 + source_video.parent.mkdir(parents=True)
744 + out.mkdir(parents=True)
745 + traj.write_text("x\n", encoding="utf-8")
746 + labels.write_text("start_time,end_time\n0,1\n", encoding="utf-8")
747 + source_video.write_text("video\n", encoding="utf-8")
748 + manifest = out / "human_lovo_manifest.json"
749 + manifest.write_text(
750 +     json.dumps(
751 +         [
752 +             {
753 +                 "name": "foo",
754 +                 "trajectory": str(traj),
755 +                 "labels": str(labels),
756 +                 "fps": 30.0,
757 +                 "frame_width": 1920.0,
758 +             }
759 +         ],
760 +     ),
761 +     encoding="utf-8",
762 + )
763 + source_manifest = workspace / "golden_source_videos.json"
764 + source_manifest.write_text(
765 +     json.dumps(
766 +         {
767 +             "schema": "sports-data-collector.golden_source_videos.v2",
768 +             "videos": [
769 +                 {
770 +                     "video_id": "foo",
771 +                     "local_path": str(source_video),
772 +                     "source_status": "present_local",
773 +                 }
774 +             ],
775 +         },
776 +     ),
777 +     encoding="utf-8",
778 + )
779 + feature_json = out / "foo_golden_features.json"
780 + feature_json.write_text(
781 +     json.dumps(
782 +         {
783 +             "name": "foo",
784 +             "fps": 30.0,
785 +             "frame_width": 1920.0,
786 +             "candidates": [{"start": 0.0, "end": 1.0}],
787 +             "gts": [[0.0, 1.0]],
788 +         },
789 +     ),
790 +     encoding="utf-8",
791 + )
792 + seen = []
793 +
794 + def _fake_detect(video_path, k=2.5):
795 +     seen.append((video_path, k))
796 +     return [H(0.5, 3.0)]
797 +
798 + monkeypatch.setattr(fusion_audio, "detect_hits_in_video", _fake_detect)
799 +
800 + written = fusion_audio.augment(
801 +     str(manifest), str(out), k=1.5, source_manifest=str(source_manifest)
802 + )
803 +

```

```

804 +     assert written == [str(feature_json)]
805 +     assert seen == [(str(source_video), 1.5)]
806 +
807 +
808     def test_audio_increment_compare(tmp_path):
809         d = str(tmp_path)
810         for k in range(6):
811             diff --git a/tests/test_fusion_compare.py b/tests/test_fusion_compare.py
812             index f042eea1..6b4226e6 100644
813             --- a/tests/test_fusion_compare.py
814             +++ b/tests/test_fusion_compare.py
815             @@ -181,6 +181,7 @@ def detections_over_windows(self, video, windows, frame_skip,
progress_cb=None):
816                 "fps": 30.0,
817                 "frame_width": 1920.0,
818                 "labels": "foo.rallies.csv",
819 +                 "video_path": __file__,
820             }
821             data = fusion_golden.extract_video(
822                 spec, detector=_StubDetector(), frame_skip=3, iou=0.5
823             @@ -197,6 +198,47 @@ def detections_over_windows(self, video, windows, frame_skip,
progress_cb=None):
824                 assert all(isinstance(v, float) for v in c0["person"].values())
825
826
827 +def test_extract_video_aborts_on_max_gpu_temp(monkeypatch):
828 +    from backend.eval import fusion_golden
829 +    from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
830 +
831 +    monkeypatch.setattr(
832 +        fusion_golden.distill_local,
833 +        "build_candidates",
834 +        lambda traj, fps, fw: ([[(0.0, 1.0)], [[1.0, 0.1, 0.1, 0.1, 0.0]]]),
835 +    )
836 +    monkeypatch.setattr(
837 +        TrackNetRunner, "parse_trajectory_csv", staticmethod(lambda traj: [])
838 +    )
839 +    monkeypatch.setattr(fusion_golden, "load_golden_gts", lambda labels: [])
840 +
841 +    class _HotTelemetry:
842 +        def snapshot(self, done, total, phase=None):
843 +            return {"gpu": {"temp_c": 90.0}}
844 +
845 +    class _StubDetector:
846 +        def detections_over_windows(self, video, windows, frame_skip,
progress_cb=None):
847 +            progress_cb(1, 10)
848 +            return []
849 +
850 +    spec = {
851 +        "name": "hot",
852 +        "trajectory": "bar.csv",
853 +        "fps": 30.0,
854 +        "frame_width": 1920.0,
855 +        "labels": "foo.rallies.csv",
856 +        "video_path": __file__,
857 +    }
858 +
859 +    with pytest.raises(RuntimeError, match="GPU temperature 90.0C"):
860 +        fusion_golden.extract_video(
861 +            spec,
862 +            detector=_StubDetector(),
863 +            telemetry=_HotTelemetry(),
864 +            max_gpu_temp_c=85.0,
865 +        )

```

```

866 +
867 +
868 def test_fusion_golden_run_is_resumable(tmp_path, monkeypatch):
869     """run() skips a video whose JSON already exists (resume a long run); --fresh
redoes all."""
870     from backend.eval import fusion_golden
871     @@ -261,6 +303,121 @@ def _stub(spec, **kw):
872         assert extracted == ["alpha", "beta"] # --fresh redoes both
873
874
875 +def test_fusion_golden_run_reextracts_windowing_only_json(tmp_path, monkeypatch):
876 +    """A7: an existing lean windowing JSON is stale for fusion and must be
regenerated."""
877 +    from backend.eval import fusion_golden
878 +
879 +    out_dir = str(tmp_path / "out")
880 +    os.makedirs(out_dir)
881 +    (tmp_path / "a.csv").write_text("x\n", encoding="utf-8")
882 +    (tmp_path / "a.rallies.csv").write_text("x\n", encoding="utf-8")
883 +    manifest = [
884 +        {
885 +            "name": "alpha",
886 +            "trajectory": "a.csv",
887 +            "fps": 30.0,
888 +            "frame_width": 1920.0,
889 +            "labels": "a.rallies.csv",
890 +        }
891 +    ]
892 +    manifest_path = str(tmp_path / "m.json")
893 +    with open(manifest_path, "w", encoding="utf-8") as f:
894 +        json.dump(manifest, f)
895 +    _write_windowing_video(out_dir, "alpha")
896 +
897 +    extracted = []
898 +
899 +    def _stub(spec, **kw):
900 +        extracted.append(spec["name"])
901 +        return {
902 +            "name": spec["name"],
903 +            "fps": 30.0,
904 +            "frame_width": 1920.0,
905 +            "candidates": [],
906 +            "gts": [],
907 +        }
908 +
909 +    monkeypatch.setattr(fusion_golden, "extract_video", _stub)
910 +
911 +    fusion_golden.run(manifest_path, out_dir, fresh=False)
912 +
913 +    assert extracted == ["alpha"]
914 +    assert fusion_golden.is_full_fusion_feature_file(
915 +        os.path.join(out_dir, "alpha_golden_features.json")
916 +    )
917 +
918 +
919 +def test_fusion_golden_source_manifest_supplies_video_paths(tmp_path, monkeypatch):
920 +    """The collector source manifest can provide Drive/F-drive video paths for A7
extraction."""
921 +    from backend.eval import fusion_golden
922 +
923 +    out_dir = str(tmp_path / "out")
924 +    os.makedirs(out_dir)
925 +    for name in ("a.csv", "b.csv", "a.rallies.csv", "b.rallies.csv"):
926 +        (tmp_path / name).write_text("x\n", encoding="utf-8")
927 +    source_video = tmp_path / "source-alpha.mp4"

```

```

928 +     source_video.write_text("video\n", encoding="utf-8")
929 +     source_manifest = tmp_path / "golden_source_videos.json"
930 +     source_manifest.write_text(
931 +         json.dumps(
932 +             {
933 +                 "schema": "sports-data-collector.golden_source_videos.v2",
934 +                 "videos": [
935 +                     {
936 +                         "video_id": "alpha",
937 +                         "local_path": str(source_video),
938 +                         "source_status": "present_local",
939 +                     },
940 +                     {
941 +                         "video_id": "beta",
942 +                         "local_path": str(tmp_path / "missing.mp4"),
943 +                         "source_status": "missing_local_source_hash_from_run_log",
944 +                     },
945 +                 ],
946 +             }
947 +         ),
948 +         encoding="utf-8",
949 +     )
950 +     manifest = [
951 +         {
952 +             "name": "alpha",
953 +             "trajectory": "a.csv",
954 +             "fps": 30.0,
955 +             "frame_width": 1920.0,
956 +             "labels": "a.rallies.csv",
957 +             "video_path": str(tmp_path / "stale-alpha.mp4"),
958 +         },
959 +         {
960 +             "name": "beta",
961 +             "trajectory": "b.csv",
962 +             "fps": 30.0,
963 +             "frame_width": 1920.0,
964 +             "labels": "b.rallies.csv",
965 +         },
966 +     ]
967 +     manifest_path = str(tmp_path / "m.json")
968 +     with open(manifest_path, "w", encoding="utf-8") as f:
969 +         json.dump(manifest, f)
970 +
971 +     seen = []
972 +
973 +     def _stub(spec, **kw):
974 +         seen.append((spec["name"], spec.get("video_path")))
975 +         return {
976 +             "name": spec["name"],
977 +             "fps": 30.0,
978 +             "frame_width": 1920.0,
979 +             "candidates": [],
980 +             "gts": [],
981 +         }
982 +
983 +     monkeypatch.setattr(fusion_golden, "extract_video", _stub)
984 +
985 +     fusion_golden.run(manifest_path, out_dir, source_manifest=str(source_manifest))
986 +
987 +     assert seen == [("alpha", str(source_video)), ("beta", None)]
988 +
989 +
990 + def test_fusion_golden_smallest_first(tmp_path, monkeypatch):
991 +     """--smallest-first orders by trajectory size; default keeps manifest order."""
992 +     from backend.eval import fusion_golden

```

8. user (2026-07-04T00:45:24.914Z)

```

1      # CLAUDE.md ? project memory / agent handoff
2
3      Local, AI-powered badminton (? racket-sports) highlight indexer + rally detector**:
4      FastAPI + SQLite + ffmpeg + GPU CV/AI, with paid Gemini as the cloud AI. Sibling
5      repos: `sports-data-collector`, a VLC-based `rally-annotator`.
6
7      ## Read these first
8      - docs/README.md ? doc index. docs/CODE_MAP.md ? code navigation + data
9        contracts (the cold-start map; read before touching code).
10     - Authoritative owned-model roadmap: docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md (+
the
11     live docs/NEXT_STEPS.md). These SUPERSEDE the 2026-06-07 strategy docs below where
they
12     disagree (licensing, Gemma-4 stance, base model, conversational-QA).
13     - Strategy (post-scaffolding): docs/COMMERCIALIZATION.md,
docs/PMF_MARKET_RESEARCH.md,
14     docs/PLATFORM_ARCHITECTURE.md, docs/DEFERRED_HARDENING_PLAN.md. Also
docs/ROADMAP.md
15     (offline-segmenter narrative) and docs/BACKLOG.md.
16     - Rally-detection quality track (design merged 2026-06-13, owner-gated, implementation
in progress):
17     docs/HOW_RALLY_DETECTION_WORKS.md (2-layer mental model) ?
docs/MULTI_SIGNAL_FUSION_PLAN.md
18     (fuse shuttle + person + audio cues; the held-shuttle bug is the already-built
rally_gate.py
19     "Gate A" ? implemented in backend/eval/served_gate_a.py, default-OFF) ?
docs/EXPERIMENT_HARNESS.md (A/B + shadow
20     eval so cues land flag-gated/default-OFF with zero regression) ?
docs/ROORA_LABELING_GUIDELINES.md
21     (v8 vendor labeling spec). docs/NEXT_STEPS.md has the prioritized pick-up for this
track.**
22     - History: docs/archives/ ? ADRs (decisions/DECISIONS.md), research, and
23     checkpoints/ (latest: 2026-06-strategy-foundation/). Archive policy: only
move
24     terminal-state (DONE/REJECTED) work; live docs stay at docs/ top level ? and
before
25     archiving, HONESTLY update the doc first (verify claims vs code ? flip status + a
completion
26     note ? update inbound refs ? keep the lineage ? then git mv), per the checklist +
living
27     lifecycle register in docs/DOC_STATUS.md.
28
29     ## Current state (July 2026)
30     - Platform slice SHIPPED: async job model (backend/api/job_worker.py, #339),
31     StorageBackend + ComputeBackend registries with concrete implementations
32     (backend/storage/ ? local/gcs/gdrive/gdrive_api/s3; backend/compute/ ?
33     local/cloud_run), capabilities/assets endpoints, and the data flywheel
34     (backend/flywheel.py). COMPUTE_DECOUPLED_SERVING M0?M9 largely shipped
35     (ADR-014 ratified 2026-06-21); Gen-0 owned-model weights pushed (weights/0.1.0-l4/).
36     - Launch decision: NO-GO (owner, 2026-06-30, #404) ? the rally-detection product is
37     NOT cleared for external/commercial launch; we stay in development. Two unblockers
gate a
38     future GO: (1) set + meet the ship-gate target (#172, needs corpus growth); (2)
resolve
39     WASB shuttle-detector weight provenance (commercial-clean licensing).
40     - Active tracks (see docs/NEXT_STEPS.md for the prioritized pending list):
41     rally-quality / owned-model ML work (gated on golden-corpus growth), packaging
42     (docs/PACKAGING_PLAN.md), and the khelsutra SPA/deploy sibling track.
43
44     ## Architecture in one breath
45     - Pluggable registries (ABC + Registry singleton): segmenters / providers /

```

```

validators
46      / sports / annotations ? see `backend/pipeline/segmenters/base.py`.
47      `StorageBackend` + `ComputeBackend` registries are implemented
(`backend/storage/`,
48      `backend/compute/`; PLATFORM §3 ? realized by `COMPUTE_DECOUPLED_SERVING`).
49      - `video_id` = MD5 of the file ? `backend/utils/hashing.py::compute_video_id` (one
helper).
50      - Typed config = `backend/config/` (Pydantic). DB schema evolves via the `PRAGMA
51      user_version` migration ladder in `backend/infrastructure/database.py`.
52
53      ## Committed guardrails (non-negotiable)
54      - Paid LLMs (Gemini/OpenAI/Anthropic) = inference / serving / fallback ONLY ? never a
55      training data source (terms/copyright). Commercial-clean licensing for EVERY
dependency ?
56      code, data, AND weights: MIT / Apache-2.0 / BSD only (ratified 2026-06-09, owner-
agreed).
57      - Owned-model base: Qwen3-VL (Apache-2.0) primary, InternVL3 (MIT) fallback;
Gemma-4
58      is AMBER / bench-only (Apache grant likely but Prohibited-Use posture unconfirmed ?
counsel
59      needed; do NOT build on it without sign-off). Reject Gemma 3 (viral license).
Exclude
60      minors from the training pool; per-user training data needs a separate explicit opt-
in.
61      - v1 = rally + highlight + history + correction loop; gait/biometrics strictly
future
62      (GDPR Art. 9).
63      - Generic data schema; `coach ? athlete` is an optional overlay, not baked in.
64      - Compute = GCP ONLY (ADR-013, 2026-06-20): GPU training + serving run on GCP
(`deploy/gcp/`;
65      Cloud Run+GPU for serving per ADR-012). DigitalOcean is DROPPED for compute (DO
GPU not enabled
66      on the account + not cleanly credit-funded; Paperspace is subscription-gated). Don't
re-evaluate
67      DO/Paperspace for compute without a new ADR; the `$200` DO credits go to non-compute
only. The
68      `deploy/digitalocean/` scripts are kept solely as provider-agnostic Linux-GPU setup
reused by GCP.
69
70      ## Workflow conventions
71      - Branch off `master`; ONE PR per work item; NO PR stacking. Don't open a PR unless
asked.
72      - Tests: `python run_all_tests.py` (offline, fully mocked). Python 3.10+ (the `trackers`
package requires ?3.10); `ffmpeg` required.
73      - Never push to the default branch directly; descriptive commit messages.
74      - Substantial design/project work gets a tracked project doc following
`docs/PROJECT_DOC_TEMPLATE.md` (status header · decisions-locked · progress tracker ·
definition-of-done; archived on completion). Precedents: `docs/RALLY_QUALITY_RESEARCH.md` §7,
`docs/GOLDEN_REGRESSION_FIXTURES.md`.
75      - Launch / "bit-flip" PRs (a default-flip that drives a tracked project to terminal
state) MUST, in the SAME PR, run the `docs/DOC_STATUS.md` §2 archival due-diligence on that
project's docs (verify-vs-code ? flip status + completion note ? update inbound refs ? keep ADR
lineage ? `git mv` to `docs/archives/` ? record in DOC_STATUS §3), and fix any internal-link
breakage the move causes. Enforced by `.github/PULL_REQUEST_TEMPLATE.md` + the `link-check` CI
job (`python -m tools.check_md_links`).
76      - ?? Hand the heaviest machine-side checks (real video, the full GPU/WSL/WASB suite) to
the owner,
77      who has the golden corpus + labelled videos; verify pure-logic paths locally.
78

```

9. user (2026-07-04T00:45:25.098Z)

```

1      # Alpha Launch Readiness Plan
2
3      > Status: `DRAFT` - cross-repo execution tracker; Owner: @avidullu; Created:

```



```

2026-07-03; **Last updated:** 2026-07-04
4     > **Lifecycle:** `DRAFT -> IN PROGRESS -> DONE -> archived`
5     > **Tracking anchors:** Section 7 is the progress tracker and source of truth; indexed
in `docs/README.md` and `docs/DOC_STATUS.md`.
6     > **Relation to existing docs:** consolidates the launch-relevant pending items from
`NEXT_STEPS.md`, `ROADMAP.md`, `docs/data_pipeline/GOLDEN_VIDEOS.md`,
`R2_EVAL_METRIC_DESIGN.md`, `COMPUTE_DECOUPLED_SERVING/`, and the sibling `kheIsutra` Studio
docs.
7     > **Honesty note:** `[verified]` means checked against the local synced repos on
2026-07-03. `[analysis]` means derived from a local command run on that snapshot. This is not
legal, financial, or launch approval.
8
9     ---
10
11     ## 0. TL;DR
12
13     The corpus is now large enough to move from "grow labels first" into alpha-readiness
work, but the 15-video headline is not itself a launch gate pass. `[verified]` The golden
manifest has 15 badminton videos; `[analysis]` A2-A7 made the local manifest/path and full-
fusion/audio feature surface reproducible on all 15; `[analysis]` A4/A5 refreshed the n=15
nightly and heuristic floors; `[analysis]` A8/A9 still reject served Gate A default promotion
and owned-model serving claims on the current evidence.
14
15     The first controlled alpha should be a hosted, authenticated product loop: upload a
video, process it, select rallies, compile, and download a reel. The owned-model/commercial
rally-detection claim remains a separate NO-GO until the ship-gate target is ratified and the
WASB checkpoint provenance is resolved.
16
17     ## 1. Problem & goal
18
19     The previous launch blockers were spread across roadmap docs and mixed together three
different questions:
20
21     - **Can an invited tester use the product without SSH?**
22     - **Can the 15-video golden corpus now support honest quality gates?**
23     - **Can we claim or ship an owned/commercial rally detector?**
24
25     This doc separates those into executable workstreams. The goal is to reach an invite-
only alpha where the product is usable end-to-end, the quality claims are honest and
reproducible, and the remaining model/legal gates are explicit rather than accidentally implied
by launch activity.
26
27     ## 2. Decisions locked
28
29     | # | Decision | Source / date | Implication |
30     |---|---|---|---|
31     | D1 | **Product alpha and owned-model launch are separate decisions.** |
`NEXT_STEPS.md` NO-GO banner, 2026-06-30; verified 2026-07-03 | We can launch a controlled
hosted product loop without claiming the owned model is commercially cleared. |
32     | D2 | **First alpha must be page-driven, not SSH/CLI-driven.** | `NEXT_STEPS.md` P0 UI-
driven/shareable item; verified 2026-07-03 | Hosted service, edge auth, upload, job polling,
compile, and download are launch-critical. |
33     | D3 | **The 15-video corpus is now a measurement asset, not a free launch pass.** |
`[verified]` `docs/data_pipeline/GOLDEN_VIDEOS.md`; `[analysis]` local eval runs, 2026-07-03 |
Rebase/portability, feature completeness, baseline refresh, and gate ratification come before
quality claims. |
34     | D4 | **Do not promote served Gate A by default from the current n=15 evidence.** |
`[analysis]` `backend.eval.serve_contrast` run, 2026-07-04 | Proposal-side lift does not survive
served-windowing safety; keep it as an opt-in/analysis lever until owner ratification. |
35     | D5 | **P10/P11 corpus promotion and train/re-eval are alpha-plus unless owner reopens
D13.** | sibling `kheIsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`; verified 2026-07-03 | First alpha
can use manual/offline corpus promotion; automated flywheel follows after serving and gates are
stable. |
36     | D6 | **Gen-0 n=15 shadow training is non-serving.** | `[analysis]`
`training.gen0.harness` run, 2026-07-04 | Learned mean beats the heuristic floor, but the paired

```

```

sign test is not significant; #172 and WASB C3 remain hard blockers. |
37
38     ## 3. Verified snapshot, 2026-07-03
39
40     All five Git checkouts were fast-forwarded before analysis:
41
42     | Repo | Branch | Synced HEAD |
43     |---|---|---|
44     | `badminton-highlight-indexer` | `master` | `e4914d1` |
45     | `khelsutra` | `master` | `b0f70a5` |
46     | `rally-annotator` | `main` | `ef9e088` |
47     | `sports-data-collector` | `master` | `bfc014f` |
48     | `sports-obsreport` | `main` | `a0c7252` |
49
50     Corpus/eval facts:
51
52     - `[verified]` `docs/data_pipeline/GOLDEN_VIDEOS.md` says
53     `output/human_lovo_manifest.json` is the ingested golden corpus at `n = 15`.
54     - `[analysis]` The local manifest had stale absolute paths. A temporary rebased manifest
55     found all 15 trajectories and labels in this workspace.
56     - `[analysis]` A7 generated full shuttle/person fusion schema for all 15 videos and A8
57     audio augmentation populated audio features for all 15.
58     - `[analysis]` `python -m backend.eval.fusion_compare --features-dir output` now runs
59     over all 15 and measures person-fusion `Delta F1 = +0.008`; the paired CI crosses zero, so
60     person fusion is not promotable by itself.
61     - `[analysis]` `python -m backend.eval.fusion_audio --features-dir output --compare`
62     measures audio `Delta F1 = +0.018`; the paired CI crosses zero, so audio is a shadow/suggestive
63     cue, not a default-flip.
64     - `[analysis]` `python -m backend.eval.serve_contrast --manifest
65     output/human_lovo_manifest.json` measured proposal Gate A mean `Delta F1 +0.0287`, but served
66     mean `Delta F1 -0.0328`; not served-safe.
67     - `[analysis]` `python -m training.gen0.harness --manifest
68     output/human_lovo_manifest.json --floor eval_baselines/heuristic_lovo_n15.json --version
69     0.1.1-n15-shadow.20260704 --out artifacts/rally_tal_gen0` produced learned LOVO mean F1 `0.551`,
70     floor passed versus heuristic `0.446`, but sign test `9/15`, `p = 0.3036`, `ship=False`.
71     - `[analysis]` `python scripts/nightly_regression.py --manifest <rebased-n15>
72     --features-dir output --no-report` reported default/milestone tolF1 `0.3636` vs baseline-off
73     tolF1 `0.1382`, but marked the frozen baseline stale due to config/corpus changes.
74     - `[analysis]` Before A6, `python -m backend.eval.ablation --features-dir output
75     --granularity group` failed at import due to a circular import between
76     `backend/eval/ablation.py` and `backend/eval/ablation_pdf.py`; A6 fixes the module entrypoint
77     and keeps full-fusion ablation scoped to the 6 videos with full schema until A7.
78
79     ## 4. Launch strategy
80
81     ### 4.1 Alpha lane: make the product usable
82
83     The alpha lane is successful when an invited tester can open the app, authenticate,
84     upload a video, watch the job progress, pick rallies, compile a reel, and download it. This lane
85     should avoid model-quality promises beyond the measured state and should default to the most
86     reliable serving path.
87
88     ### 4.2 Quality lane: make the 15-video corpus gate-ready
89
90     The quality lane is successful when the 15-video corpus is portable, reproducible,
91     feature-complete, and backed by refreshed baselines. Only then should R2 tolF1,
92     Gen-0/TemporalMaxer, or other model/promotion decisions be ratified.
93
94     ### 4.3 Flywheel lane: make tester feedback useful
95
96     The flywheel lane is successful when user corrections can flow into the collector/vault
97     path with consent/provenance and a visible re-eval delta. This is valuable for alpha learning,
98     but it should not block the first controlled alpha if manual promotion can bridge the gap.
99
100    ## 5. Risks and guardrails

```

```

77
78     | Risk | Why it matters | Guardrail |
79     |---|---|---|
80     | Launch activity implies model launch approval | The engine docs explicitly say rally-
detection product launch is NO-GO until ship target and WASB C3 clear | Keep all alpha copy
scoped to invite-only processing, not owned-model claims |
81     | Stale local manifests create false confidence | Manifest-driven tools skipped or
failed until paths were rebased | Make the corpus manifest portable and add a path/fixture smoke
check |
82     | 15-video corpus is only partially feature-complete | Fusion/person/audio eval is still
n=6 today | Regenerate or upconvert full-fusion features for all 15 before fusion decisions |
83     | Baseline refresh accidentally rubber-stamps a regression | Nightly runner correctly
reported stale baselines | Refresh baselines only after reviewer sign-off and record the
command/output |
84     | P10/P11 expands scope before hosted alpha is usable | Corpus flywheel is valuable but
can consume the alpha launch window | Keep P10/P11 behind explicit owner reopen unless alpha
lane is already stable |
85     | WASB checkpoint provenance remains unresolved | Blocks commercial sharing of a trained
model regardless of F1 | Treat C3 as a hard model-release gate; do not hide it inside product
launch tasks |
86
87     ## 6. Honest limits
88
89     Completing this doc's alpha lane does **not** prove the owned model is ready, that WASB
weights are commercially clean, that the product can serve unbounded public traffic, that multi-
sport is ready, or that every tester correction automatically improves the model.
90
91     Completing the quality lane does **not** by itself deploy a service. It only makes the
measurement surface honest enough for a launch decision.
92
93     ## 7. Deliverables & progress tracker
94
95     Legend: TODO = not started, IN PROGRESS = actively being worked, DONE =
shipped/verified, GATED = blocked on owner/legal/platform decision. One PR per row unless the
row explicitly names a cross-repo batch.
96
97     | ID | Deliverable | Repo / owner | Depends on | Gated? | Status | PR |
98     |---|-----|-----|-----|-----|-----|---|
99     | A0 | Create this alpha readiness tracker and index it in docs | `badminton-highlight-
indexer` | - | No | DONE | #451 |
100    | A1 | Stand up alpha serving endpoint: Cloud Run/GPU or approved host, health check,
edge auth ON, upload -> process -> jobs -> compile -> download verified | `khelsutra` +
`badminton-highlight-indexer` | A0 | Owner spend / CF config | TODO | - |
101    | A2 | Make the n=15 golden manifest portable/reproducible; add a path/corpus smoke
check that fails loudly when labels or trajectories are missing | `badminton-highlight-indexer`
| A0 | No | DONE | #452 |
102    | A3 | Promote the 15-video source-video/MD5 records into the collector/vault path;
reconcile collector's six-video source manifest with the 15-video eval corpus | `sports-data-
collector` + vault | A2 | Owner upload/storage scope | IN PROGRESS | collector #62 |
103    | A4 | Refresh the n=15 nightly regression baseline intentionally and record the exact
command/output; keep stale-baseline warnings until reviewed | `badminton-highlight-indexer` | A2
| Reviewer sign-off | DONE | #453 |
104    | A5 | Recompute the heuristic n=15 floor and replace the stale `heuristic_lovo_n6`
floor for future Gen-0 comparisons | `badminton-highlight-indexer` | A2 | No | DONE | #454 |
105    | A6 | Fix `backend.eval.ablation` CLI circular import, then run the R2/toLF1 ablation
on the n=15 corpus | `badminton-highlight-indexer` | A2 | No | DONE | #455 |
106    | A7 | Regenerate or upconvert full-fusion feature JSONs for the newer 9 videos so
fusion/person/audio analyses run over all 15 | `badminton-highlight-indexer` | A2 | GPU/data
availability | IN PROGRESS | this PR |
107    | A8 | Re-run `fusion_compare`, group ablation, and served contrast after A7; explicitly
promote/reject person fusion, Gate A served-default, and any other R-series default flips |
`badminton-highlight-indexer` | A6, A7 | Reviewer sign-off | IN PROGRESS | this PR |
108    | A9 | Run Gen-0/TemporalMaxer shadow training against the refreshed n=15 floor; produce
a non-serving artifact bundle and gate verdict | `badminton-highlight-indexer` | A5, A8 | Owner
approval for M0.4 scope | IN PROGRESS | this PR |

```

```

109 | A10 | Decide and record the alpha ship-gate target: metric, corpus slice, minimum
threshold, significance rule, and what claims are allowed | Owner + `badminton-highlight-
indexer` | A4, A5, A8 | Owner decision | GATED | - |
110 | A11 | Resolve or explicitly defer WASB C3 provenance for alpha messaging; if deferred,
document "no commercial model sharing" copy guardrail | Owner/legal | A10 | Legal/provenance |
GATED | - |
111 | A12 | Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector
promotion, `train`/`reeval` job, delta-F1 surface, and alignment with
`COMPUTE_DECOUPLED_SERVING` M11 | `khelsutra` + `badminton-highlight-indexer` + collector | A1,
A3 | Owner D13 reopen (`khelsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`) | GATED | - |
112 | A13 | Alpha launch review: runbook, privacy/consent copy, observability/job-failure
triage, known limits, rollback/disable steps, and owner GO/NO-GO | Cross-repo | A1, A4, A10, A11
| Owner launch decision | TODO | - |
113
114 ## 8. Concrete next PR sequence
115
116 1. **Docs PR:** land this tracker.
117 2. **Corpus portability PR:** rebase/portable manifest tooling plus missing-file smoke
test.
118 3. **Hosted alpha PR/batch:** health/auth/upload/process/job/compile/download
verification and runbook updates.
119 4. **Baseline PR:** intentional n=15 nightly baseline refresh and heuristic floor
refresh.
120 5. **Ablation/fusion PRs:** fix ablation import, complete 15-video features, rerun
promotion/rejection decisions.
121 6. **Launch decision PR/docs update:** record ship-gate target, C3 status, alpha copy
limits, and launch checklist.
122
123 A2 smoke command:
124
125 ```powershell
126 python -m backend.eval.golden_manifest --manifest output/human_lovo_manifest.json --min-
count 15
127 python -m backend.eval.golden_manifest --manifest output/human_lovo_manifest.json --min-
count 15 --write output/human_lovo_manifest.local.json
128 ```
129
130 The checker resolves stale absolute paths against the repo's sibling `Annotation Setup`
folders and `output/`, then exits non-zero if any required `trajectory` or `labels` file is
absent.
131
132 A4 nightly baseline refresh record:
133
134 ```powershell
135 python scripts\nightly_regression.py --manifest output\human_lovo_manifest.json
--features-dir output --update-baseline
136 ```
137
138 Result:
139
140 ```text
141 INFO baseline snapshot written to C:\Users\avidu\Projects\khelsutra-guru\badminton-
highlight-indexer\eval_baselines\nightly_baseline.json
142 INFO default: n=15 tolF1=0.364 f1@0.5=0.472 merge_rate=0.135
143 INFO milestone: n=15 tolF1=0.364 f1@0.5=0.472 merge_rate=0.135
144 INFO baseline_off: n=15 tolF1=0.138 f1@0.5=0.237 merge_rate=0.638
145 ```
146
147 Pass-mode proof:
148
149 ```powershell
150 python scripts\nightly_regression.py --manifest output\human_lovo_manifest.json
--features-dir output --no-report
151 ```
152

```

```

153     Result: `PASS`, exit code `0`, baseline snapshot `2026-07-03`, default/milestone `tol_f1
= 0.3636`, `f1@0.5 = 0.4723`, `merge_rate = 0.1350`; baseline-off `tol_f1 = 0.1382`, `f1@0.5 =
0.2366`, `merge_rate = 0.6380`.
154
155     A5 heuristic floor refresh record:
156
157     ```powershell
158     python -m backend.eval.calibrate_local --manifest output\human_lovo_manifest.json
--metric f1 --floor-out eval_baselines\heuristic_lovo_n15.json
159     ```
160
161     Result:
162
163     ```text
164     baseline (untuned) mean held-out f1: 0.237
165     calibrated mean held-out f1:          0.446 +/- 0.239 (gap +0.000)
166     ```
167
168     The committed floor is `eval_baselines/heuristic_lovo_n15.json`: `n = 15`, `mean_f1 =
0.4456`, `std = 0.2390`, `metric = f1`, `iou = 0.5`. Every leave-one-video-out fold selected
`(0.05, 1.0, 2.0, 0)`.
169
170     A6 ablation/R2 record:
171
172     ```powershell
173     python -m backend.eval.ablation --features-dir output --granularity group
174     python -m backend.eval.rally_seg_eval --manifest output\human_lovo_manifest.json
--features-dir output --preset served --max-window-duration 0 --inactivity-end 0 --vs served
--vs-max-window-duration 6 --vs-inactivity-end 1.5 --json-out output\a6_r2_ablation.json
175     ```
176
177     Result: the ablation CLI now runs, but full-fusion feature ablation still reports `n=6`
because only the original six feature JSONs carry the full fusion schema; A7 must
regenerate/upconvert the newer nine before fusion/person/audio promotion decisions. The n=15
R2/tolF1 contrast is positive for the served milestone versus baseline-off: baseline/off `tolF1
= 0.138`, served `tolF1 = 0.364`, `Delta tolF1 = +0.225` with CI `[+0.165, +0.283]`, sign
`14/0`, `p = 0.000`; `Delta F1@0.5 = +0.236` with CI `[+0.161, +0.313]`; merge rate improves
from `0.638` to `0.135`. Net over-segmentation guard passes (`1.276 -> 0.378`), while the strict
segment-count rule remains blocked by increased splits, so R2/tolF1 is the supported gate for
the baseline-off to served milestone contrast rather than a blanket fusion/default-flip
approval.
178
179     A7 full-fusion/audio feature refresh record:
180
181     ```powershell
182     python -m backend.eval.fusion_golden --manifest output\human_lovo_manifest.json --out-
dir output --source-manifest ..\sports-data-collector\data\golden_source_videos.json --missing-
fusion-only --smallest-first --max-gpu-temp-c 87
183     python -m backend.eval.fusion_audio --manifest output\human_lovo_manifest.json
--features-dir output --source-manifest ..\sports-data-collector\data\golden_source_videos.json
--augment
184     ```
185
186     Result: the GPU pass completed without a thermal abort after fans were set to max
performance. Full-fusion schema is now present for `15/15` feature JSONs, and audio features are
present for `15/15`. The large regenerated clips completed as follows: `mahadevpura_1` `1`
window / `0` positives / `10` GT, `GX020094` `99` / `30` / `40`, `GX030094` `79` / `21` / `41`,
`GX010141` `26` / `11` / `29`, `GX010137` `56` / `31` / `34`, `Boxhill_Doubles` `59` / `12` /
`43`, `mahadevpura_2` `14` / `2` / `40`, `GX010128` `51` / `15` / `52`, `Badminton_BXH_2` `29` /
`1` / `44`. The source manifest supplied all 15 source-video paths, including the F-drive
Boxhill originals.
187
188     A8 fusion/default-decision record:
189
190     ```powershell

```

```

191     python -m backend.eval.fusion_compare --features-dir output
192     python -m backend.eval.fusion_audio --features-dir output --compare
193     python -m backend.eval.ablation --features-dir output --granularity group
194     python -m backend.eval.serve_contrast --manifest output\human_lovo_manifest.json
195     python -m backend.eval.rally_seg_eval --manifest output\human_lovo_manifest.json
--features-dir output --preset served --max-window-duration 0 --inactivity-end 0 --vs served
--vs-max-window-duration 6 --vs-inactivity-end 1.5 --json-out output\a6_r2_ablation.json
196     ``
197
198     Result:
199
200     - Person fusion is not promoted: shuttle-only F1 `0.302`, shuttle+person F1 `0.311`,
`Delta F1 +0.008`, paired CI `[-0.040, +0.055]`, sign `8W/6L`, `p = 0.791`.
201     - Audio is shadow/suggestive, not promoted: shuttle+person F1 `0.311`,
shuttle+person+audio F1 `0.329`, `Delta F1 +0.018`, paired CI `[-0.017, +0.053]`, sign `10W/4L`,
`p = 0.180`.
202     - Group ablation remains non-significant for cue promotion: audio `+0.018` CI `[-0.017,
+0.053]`; shuttle `+0.013` CI `[-0.002, +0.029]`; gate_a `-0.001` CI `[-0.012, +0.008]`; person
`-0.009` CI `[-0.045, +0.021]`.
203     - Gate A served default is rejected: proposal mean `Delta F1 +0.0287`, CI `[-0.0222,
+0.0680]`; served mean `Delta F1 -0.0328`, CI `[-0.0799, +0.0052]`, `regresses=True`.
204     - R2/cap/inactivity remains the strong quality result: served candidate F1 `0.472` vs
baseline-off `0.237`, `Delta F1 +0.236`, CI `[+0.161, +0.313]`; tolF1 `0.364` vs `0.138`, `Delta
tolF1 +0.225`, CI `[+0.165, +0.283]`; merge rate `0.135` vs `0.638`. Net over-seg guard passes,
strict per-video split-count guard still blocks blanket promotion language.
205
206     A9 Gen-0 shadow-training record:
207
208     ```powershell
209     python -m training.gen0.harness --manifest output\human_lovo_manifest.json --floor
eval_baselines\heuristic_lovo_n15.json --out artifacts\rally_tal_gen0 --version
0.1.1-n15-shadow.20260704
210     ``
211
212     Result: current Gen-0 LogReg/TAL harness produced local bundle
`artifacts\rally_tal_gen0\0.1.1-n15-shadow.20260704` with learned LOVO mean F1 `0.551` versus
heuristic floor `0.4456`, so `floor_passed=True`. The gate remains non-serving: paired sign
test `9/15`, `p = 0.3036`, `ship=False`; blockers are uncalibrated ship-gate target #172 and
WASB checkpoint provenance C3. TemporalMaxer itself is not implemented in this branch; the
evidence supports either closing A9 as a Gen-0 shadow baseline or opening a follow-up M0.4
scorer-swap PR.
213
214     ## 9. Open questions
215
216     1. Owner: Is the first alpha allowed to use manual/offline corpus promotion, or
should P10/P11 be reopened before any tester wave?
217     2. Owner/platform: Which hosted compute target is the next stand-up: Cloud Run GPU,
GCE L4 VM, or another approved GPU host?
218     3. Owner/legal: What is the acceptable alpha wording while WASB C3 remains
unresolved?
219     4. Engineering: Should the portable corpus manifest live as a generated artifact, a
checked-in redacted manifest, or both?
220     5. Engineering: Should A9 close on the current Gen-0 non-serving bundle, or should
the actual TemporalMaxer scorer swap be implemented before marking A9 done?
221
222     ## 10. Definition of done
223
224     - [ ] Invited tester can complete the hosted record/upload -> reel download CUJ without
SSH.
225     - [ ] Alpha service is gated behind the chosen auth posture and has a documented
disable/rollback path.
226     - [x] n=15 corpus is portable enough that manifest-driven quality commands do not rely
on stale local absolute paths.
227     - [x] n=15 nightly baseline and heuristic floor are refreshed with recorded commands.
228     - [x] R2/tolF1 gate is promoted or rejected with a recorded ablation result.

```

```

229     - [x] Fusion/person/Gate-A served-default decisions are explicitly promoted or rejected
on the 15-video corpus.
230     - [ ] Ship-gate target and allowed alpha claims are recorded.
231     - [ ] WASB C3 status is recorded and reflected in alpha messaging.
232     - [ ] P10/P11 are either reopened with a committed sequence or explicitly deferred for
alpha-plus.
233     - [ ] Owner records GO/NO-GO for the first alpha wave.
234
235     ## 11. References
236
237     Internal docs:
238
239     - `docs/NEXT_STEPS.md`
240     - `docs/ROADMAP.md`
241     - `docs/data_pipeline/GOLDEN_VIDEOS.md`
242     - `docs/R2_EVAL_METRIC_DESIGN.md`
243     - `docs/RALLY_DETECTION_FIXES_PLAN.md`
244     - `docs/COMPUTE_DECOUPLED_SERVING/README.md`
245     - sibling `khehsutra/NEXT_STEPS.md`
246     - sibling `khehsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`
247     - sibling `khehsutra/docs/PER_RALLY_SELECTION.md`
248     - sibling `sports-data-collector/data/golden_source_videos.json`
249
250     Local verification commands used for the 2026-07-03 snapshot:
251
252     ```powershell
253     python -m backend.eval.fusion_compare --features-dir output
254     python -m backend.eval.serve_contrast --manifest <rebased-n15-manifest>
255     python -m training.gen0.harness --manifest <rebased-n15-manifest> --floor
eval_baselines\heuristic_lovo_n15.json --version analysis-n15 --out <temp-out>
256     python scripts\nightly_regression.py --manifest <rebased-n15-manifest> --features-dir
output --no-report
257     python -m backend.eval.ablation --features-dir output --granularity group
258     python -m backend.eval.rally_seg_eval --manifest output\human_lovo_manifest.json
--features-dir output --preset served --max-window-duration 0 --inactivity-end 0 --vs served
--vs-max-window-duration 6 --vs-inactivity-end 1.5 --json-out output\a6_r2_ablation.json
259     ```
260
261     ---
262
263     ### Changelog
264
265     - 2026-07-03 - Created from a synced five-repo review of `NEXT_STEPS`, `ROADMAP`,
golden-corpus docs, local eval artifacts, and the current 15-video corpus state.
266     - 2026-07-04 - Recorded A7 full-fusion/audio regeneration, A8 n15 fusion/default
decisions, and A9 Gen-0 n15 shadow-training gate verdict.
267

```

10. assistant (2026-07-04T00:45:27.528Z)

Now let me check PR #455 merge status and verify any artifact references:

11. user (2026-07-04T00:45:29.756Z)

```
{"mergedAt":"2026-07-03T19:20:06Z","state":"MERGED"}
```

12. user (2026-07-04T00:45:31.116Z)

```
{"isDraft":true}
```

13. assistant (2026-07-04T00:45:35.709Z)

Perfect. Now let me analyze the violations carefully against the CLAUDE.md conventions and the doc's own honesty-tag convention stated at the top of ALPHA_LAUNCH_READINESS.md.

The key rule from global CLAUDE.md is:

> ****Keep it honest:**** it reflects real, shipped state ? never aspirational. (See each project's attribution/working-agreement note where present.)

And the honesty note in ALPHA_LAUNCH_READINESS.md states:

> ****Honesty note:**** `[verified]` means checked against the local synced repos on 2026-07-03. `[analysis]` means derived from a local command run on that snapshot. This is not legal, financial, or launch approval.

Now let me check the SESSION_HANDOFF.md changes more carefully and look for artifact reference issues:

14. user (2026-07-04T00:45:37.141Z)

ARTIFACT_EXISTS

15. assistant (2026-07-04T00:45:38.634Z)

Let me check if these artifacts are supposed to be committed:

16. user (2026-07-04T00:45:38.943Z)

```
1      # Byte-compiled / optimized / DLL files
2      __pycache__/
3      *.py[cod]
4      *$py.class
5      *.pt
6
7      # Local environment secrets
8      .env
9      *.env
10     .env.*
11     # ...but deploy/gcp/buckets.env is the TRACKED single-source-of-truth for PUBLIC project
+ bucket
12     # names (NO secrets ? same identifiers already public across the repo); the scripts
source it, so it
13     # MUST be committed. Keep real secret .env files ignored above.
14     !deploy/gcp/buckets.env
15
16     # Provenance stamp baked into a deploy tarball at build time (deploy/gcp/README.md
section 4);
17     # never a tracked source file -- generated per-deploy, must not ride into a commit.
18     /GIT_SHA
19
20     # C extensions
21     *.so
22
23     # Distribution / packaging
24     .Python
25     build/
26     develop-eggs/
27     dist/
28     downloads/
29     eggs/
30     .eggs/
31     lib/
32     lib64/
33     parts/
34     sdist/
35     var/
36     wheels/
37     share/python-wheels/
38     *.egg-info/
39     .installed.cfg
```



```

40     *.egg
41     MANIFEST
42
43     # SQLite databases & local output files
44     *.db
45     *.db-journal
46     *.db-wal
47     output/
48     output_dir/
49     backend/static/highlights_*.mp4
50     backend/static/*_compiled.mp4
51
52     # Local generated artifacts (review reels, eval CSVs, e2e/proof outputs) -- regenerable,
never committed
53     artifacts/
54
55     # OS generated files
56     .DS_Store
57     Thumbs.db
58     desktop.ini
59
60     # Coverage data
61     .coverage
62     .coverage.*
63     htmlcov/
64     .pytest_cache/
65
66     # Local Claude Code config (machine-specific)
67     .claude/
68
69     # Throwaway repro scripts + PR-body / handoff scratch notes (already excluded from
70     # ruff + pytest; never meant to be committed -- keep them out of `git status` too).
71     scratch/
72     .venv/
73
74     # Private leak-scan name-list (real player/venue names for LOCAL `tools.leak_scan
--names`
75     # scanning -- NEVER committed; CI runs the name-free structural guard).
76     .leakscan_private
77
78     # Stray scheme-prefixed report dirs from a remote URI accidentally resolved as a literal
local
79     # path (gs://b/x -> ./gs:/b/x). Root cause fixed in #397; this is a belt-and-braces
guard so a
80     # regression can never be committed. Anchored to the repo root.
81     /gs:/
82     /gs:/
83     /local:/
84     /s3:/
85     /gdrive:/
86

```

17. assistant (2026-07-04T00:45:45.179Z)

Good, `artifacts/` is explicitly ignored in the `.gitignore`. Now let me carefully review the SESSION_HANDOFF.md changes for the convention violation:

The SESSION_HANDOFF.md file:

- Line 11-12 (new): Claims "A6 PR #455 is merged"
- Line 69: References the artifact path
`artifacts/rally\_tal\_gen0/0.1.1-n15-shadow.20260704/manifest.json`

Per the global CLAUDE.md convention:

> **Keep it honest:** it reflects real, shipped state ? never aspirational.

And per the PR body claim in SESSION_HANDOFF.md lines 31-32:

> The generated feature/model artifacts live under ignored `output/` and `artifacts/` paths and should not be committed.

This creates a violation: The handoff explicitly states artifacts "should not be committed" but then lists a specific artifact path in the ramp-up kit (line 69) that will not be present when someone clones the repo or pulls this PR. According to the global CLAUDE.md session-handoff convention:

> ****Keep it honest:**** it reflects real, shipped state ? never aspirational... (See each project's attribution/working-agreement note where present.)

Let me also check the ALPHA_LAUNCH_READINESS.md for honesty-tag compliance:

18. user (2026-07-04T00:45:46.475Z)

```
7:> **Honesty note:** `[verified]` means checked against the local synced repos on 2026-07-03.
`[analysis]` means derived from a local command run on that snapshot. This is not legal,
financial, or launch approval.
13:The corpus is now large enough to move from "grow labels first" into alpha-readiness work,
but the 15-video headline is not itself a launch gate pass. `[verified]` The golden manifest has
15 badminton videos; `[analysis]` A2-A7 made the local manifest/path and full-fusion/audio
feature surface reproducible on all 15; `[analysis]` A4/A5 refreshed the n=15 nightly and
heuristic floors; `[analysis]` A8/A9 still reject served Gate A default promotion and owned-
model serving claims on the current evidence.
33:| D3 | **The 15-video corpus is now a measurement asset, not a free launch pass.** |
`[verified]` `docs/data_pipeline/GOLDEN_VIDEOS.md`; `[analysis]` local eval runs, 2026-07-03 |
Rebase/portability, feature completeness, baseline refresh, and gate ratification come before
quality claims. |
34:| D4 | **Do not promote served Gate A by default from the current n=15 evidence.** |
`[analysis]` `backend.eval.serve_contrast` run, 2026-07-04 | Proposal-side lift does not survive
served-windowing safety; keep it as an opt-in/analysis lever until owner ratification. |
36:| D6 | **Gen-0 n=15 shadow training is non-serving.** | `[analysis]` `training.gen0.harness`
run, 2026-07-04 | Learned mean beats the heuristic floor, but the paired sign test is not
significant; #172 and WASB C3 remain hard blockers. |
53:- `[analysis]` The local manifest had stale absolute paths. A temporary rebased manifest
found all 15 trajectories and labels in this workspace.
54:- `[analysis]` A7 generated full shuttle/person fusion schema for all 15 videos and A8 audio
augmentation populated audio features for all 15.
55:- `[analysis]` `python -m backend.eval.fusion_compare --features-dir output` now runs over
all 15 and measures person-fusion `Delta F1 = +0.008`; the paired CI crosses zero, so person
fusion is not promotable by itself.
56:- `[analysis]` `python -m backend.eval.fusion_audio --features-dir output --compare` measures
audio `Delta F1 = +0.018`; the paired CI crosses zero, so audio is a shadow/suggestive cue, not
a default-flip.
57:- `[analysis]` `python -m backend.eval.serve_contrast --manifest
output\human_lovo_manifest.json` measured proposal Gate A mean `Delta F1 +0.0287`, but served
mean `Delta F1 -0.0328`; not served-safe.
58:- `[analysis]` `python -m training.gen0.harness --manifest output\human_lovo_manifest.json
--floor eval_baselines\heuristic_lovo_n15.json --version 0.1.1-n15-shadow.20260704 --out
artifacts\rally_tal_gen0` produced learned LOVO mean F1 `0.551`, floor passed versus heuristic
`0.446`, but sign test `9/15`, `p = 0.3036`, `ship=False`.
59:- `[analysis]` `python scripts/nightly_regression.py --manifest <rebased-n15> --features-dir
output --no-report` reported default/milestone tolF1 `0.3636` vs baseline-off tolF1 `0.1382`,
but marked the frozen baseline stale due to config/corpus changes.
60:- `[analysis]` Before A6, `python -m backend.eval.ablation --features-dir output
--granularity group` failed at import due to a circular import between
`backend/eval/ablation.py` and `backend/eval/ablation_pdf.py`; A6 fixes the module entrypoint
and keeps full-fusion ablation scoped to the 6 videos with full schema until A7.
```

19. assistant (2026-07-04T00:45:48.309Z)

Now let me look at the PROJECT_DOC_TEMPLATE to understand the state tracking conventions:

20. user (2026-07-04T00:45:49.046Z)

```
1      <!--
2      PROJECT DOC TEMPLATE ? copy this file to docs/<YOUR_PLAN>.md and fill it in.
3
4      A "tracked project doc" is how this repo carries substantial design/project work: it can
be
5      ITERATED on, its progress TRACKED, and it is MARKED DONE against explicit deliverables,
then
6      archived. Standardizes the emergent pattern in docs/RALLY_QUALITY_RESEARCH.md §7 and
7      docs/GOLDEN_REGRESSION_FIXTURES.md.
8
9      Rules of thumb:
10     - §7 (the progress tracker) is the SOURCE OF TRUTH ? one row per independently-shippable
PR.
11     - Keep it HONEST: tag each claim [verified] (checked against code), [researched] (needs
sign-off),
12     or [design] (proposed). The doc reflects real shipped state, never aspirational.
13     - It POINTS to detail (code anchors, research artifacts), it does not duplicate it.
14     - Sections marked OPTIONAL are included only when relevant (e.g. Foundation/Threat-model
for
15     research- or security/privacy-heavy work). Delete sections that do not apply.
16     - Move to docs/archives/ once Status = DONE (per the archive policy: only terminal-state
work).
17     -->
18
19     # <Project / Plan Title>
20
21     > **Status:** `DRAFT` ? `<owner-gated? | in progress>` . **Owner:** `<name>` .
**Created:** `<YYYY-MM-DD>` . **Last updated:** `<YYYY-MM-DD>`
22     > **Lifecycle:** `DRAFT` ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
23     > **Tracking anchors:** §7 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in memory `current-state` (and `docs/NEXT_STEPS.md` once work starts).
24     > **Relation to existing docs:** `<extends | supersedes | peer-of ?>`
25     > **Honesty note:** mark each claim `[verified]` / `[researched]` / `[design]`. Not
legal/financial advice where applicable.
26
27     ---
28
29     ## 0. TL;DR
30     <2?5 sentences: the goal, the approach, and the single most important caveat. Call out
any finding that is already actionable on its own.>
31
32     ## 1. Problem & goal
33     <Why now; the concrete user-facing outcome; what "good" looks like. Link the code/docs a
reader must read to get current.>
34
35     ## 2. Decisions locked
36     <Table ? capture answered questions so they are not relitigated.>
37
38     | # | Decision | Source / date | Implication |
39     |---|-----|-----|-----|
40     | D1 | ? | ? | ? |
41
42     ## 3. Foundation ? research / prior art *(OPTIONAL)*
43     <Include when the plan rests on external facts: cited research, legal/compliance,
benchmarks, prior-art survey. Mark researched-vs-verified; list any sign-off gates this
creates.>
44
45     ## 4. Design / architecture
46     <The actual plan. Data contracts, diagrams, and a REUSE MAP ? name the existing
modules/code this builds on so it is clear what is new vs reused.>
47
48     ## 5. Threat model / risk table *(OPTIONAL ? security / privacy / compliance work)*
```

```

49     <Who/what is protected vs NOT; residual risks stated plainly.>
50
51     ## 6. Honest limits ? what this does NOT do
52     <The false-confidence guardrail: what a green suite / a DONE status does NOT prove.>
53
54     ## 7. Deliverables & progress tracker    ? **source of truth**
55     Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One small PR per row**,
56     branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.
57
58     | ID | Deliverable | Depends on | Gated? | Status | PR |
59     |----|-----|-----|-----|-----|----|
60     | P0 | ? | ? | No | ? | ? |
61
62     ## 8. Open questions ? owner / external *(split blocking-gates from nice-to-knows; name
63     who decides)*
64     1. ?
65
66     ## 9. Definition of done
67     <Explicit acceptance checklist. When all ticked ? flip `Status ? DONE` and archive.>
68     - [ ] ?
69
70     ## 10. References
71     **Internal code anchors `[verified]`:** `<paths>`. **Internal docs:** `<docs>`.
72     **External:** `<cited sources>`.
73
74     ---
75
76     ### Changelog
77     - `<YYYY-MM-DD>` ? `<what changed>`.
78
79

```

21. assistant (2026-07-04T00:46:04.023Z)

Now let me trace through the exact violations:

Violation 1: SESSION_HANDOFF.md artifact reference in ramp-up kit

```

**File:** `SESSION_HANDOFF.md`, line 69 (in diff)
**Rule violated:** Global CLAUDE.md session-handoff convention: "***Keep it honest:** it reflects
real, shipped state ? never aspirational."

```

The handoff lists in the ramp-up kit:

```

...
- `artifacts/rally_tal_gen0/0.1.1-n15-shadow.20260704/manifest.json`
...

```

But the PR explicitly states (lines 31-32):

```

> The generated feature/model artifacts live under ignored `output/` and `artifacts/` paths and
should not be committed.

```

This is a **broken reference per doc conventions** ? the artifact path is listed as something someone should read to ramp up, but it won't exist in the cloned/pulled repo because it's in `.gitignore` and not committed.

Violation 2: ALPHA_LAUNCH_READINESS.md ? "A6 PR #455 is merged" in DRAFT PR

```

**File:** `docs/ALPHA_LAUNCH_READINESS.md`, rows A6/A7/A8/A9 (lines 549-552 in diff)
**Rule violated:** PROJECT_DOC_TEMPLATE $7 legend: "One small PR per row... Update the row
(status + PR link) as work lands."

```

The rows show:

```

...
| A6 | ... | DONE | #455 |
| A7 | ... | IN PROGRESS | this PR |

```

```
| A8 | ... | IN PROGRESS | this PR |
| A9 | ... | IN PROGRESS | this PR |
...

```

This is a ****status inconsistency****: A6 is marked DONE with PR #455 (which IS merged ? verified above). But the handoff docstring says "this checkout is on `codex/a7-a9-quality-chain`... with tracked edits in..." and lists only A7/A8/A9 changes. However, the document then later references A6 results as historical fact (`[analysis]` from 2026-07-03). This is internally consistent but the tracker row for A6 in the table claiming PR #455 is merged, while the PR being reviewed (#458) is still DRAFT and IN PROGRESS, may imply a shipping state that hasn't happened yet for A7/A8/A9.

Actually, looking more carefully: A6 is correctly marked DONE #455 (merged). A7/A8/A9 are marked IN PROGRESS this PR. That's correct because PR #458 is a DRAFT PR containing A7/A8/A9 work.

Let me re-examine for actual violations:

Re-check: ALPHA_LAUNCH_READINESS.md sections with [analysis] claims

Looking at the new sections in the diff (lines 560-593):

These are all properly tagged `[analysis]` and come with the exact commands and outputs. The artifact paths in the A9 section (line 593) reference
`artifacts\rally\_tal\_gen0\0.1.1-n15-shadow.20260704` which exists locally but won't be in a committed state.

The key issue is: ****are these artifact references meant to be checked out by the next reader, or are they just historical documentation of a local run?****

Looking at the honesty note again:

> `[analysis]` means derived from a local command run on that snapshot.

So `[analysis]` claims are expected to reference local artifacts that existed at run time. The problem is in SESSION_HANDOFF.md specifically, which claims to reflect "real, shipped state" and lists specific artifact paths in the "Ramp-up kit" section ? those should be reliably available.

Let me finalize the violations:

```
```json
[
 {
 "file": "SESSION_HANDOFF.md",
 "line": 69,
 "summary": "Ramp-up kit references artifact path that won't exist in checked-out repo",
 "failure_scenario": "Reader pulls PR #458 and tries to follow ramp-up path
`artifacts/rally_tal_gen0/0.1.1-n15-shadow.20260704/manifest.json`; file does not exist because
`artifacts/` is in .gitignore and not committed. Per global CLAUDE.md session-handoff
convention: 'Keep it honest: it reflects real, shipped state ? never aspirational.'"
 }
]
```

```

22. user (2026-07-04T00:46:05.658Z)

0

23. assistant (2026-07-04T00:46:08.612Z)

Confirmed: `artifacts/` is not committed to git.

```
```json
[
 {
 "file": "SESSION_HANDOFF.md",

```

```
 "line": 69,
 "summary": "Ramp-up kit references artifact path that won't exist in checked-out repo",
 "failure_scenario": "Reader pulls PR #458 and tries to follow ramp-up path
`artifacts/rally_tal_gen0/0.1.1-n15-shadow.20260704/manifest.json`. File will not exist because
`artifacts/` is in .gitignore (line 53) and contains no committed files. Global CLAUDE.md
session-handoff rule: 'Keep it honest: it reflects real, shipped state ? never aspirational.'
This is a broken reference per PROJECT_DOC_TEMPLATE conventions on aspirational paths."
 }
]
...
```